Queryable Gaussian View Index: Retrieval, Localization, and Agentic View Search

Tianyu Xiong[1] Rui Li[2] Jiaqi Yang[1]

[1]School of Computer Science, Northwestern Polytechnical University
Xi'an, China

[2]CEMSE, King Abdullah University of Science and Technology
Thuwal, Saudi Arabia



Figure 1. **Queryable Gaussian View Index (QGVI).** QGVI-T returns caption-ranked views and stored poses (left). QGVI-I gates geometric LoGS poses (middle); the plot shows the accepted pose on SfM points. QGVI-A uses posed candidates and camera exploration to satisfy an image-relative instruction (right). Selected cached synthetic-scene observations omit intermediate proposals; descriptions are paraphrased and semantic checks are model assessments.

Abstract

Queryable Gaussian View Index (QGVI) queries existing Gaussian assets using text, RGB, or image-and-instruction inputs. QGVI-T retrieves captioned views with known poses using BM25; grid renders are optional. QGVI-I applies a distinct-keypoint acceptance gate to LoGS geometric poses. QGVI-A coordinates these interfaces: a vision-language controller checks supplied predicates, proposes camera changes, and inspects new renders until it can answer or exhausts its budget. Directly supported answers exit early. On nine development scenes, QGVI-I achieves 97.3% scene-mean pose success at 1 m/2°, with pooled false-acceptance and false-rejection rates of 0.707% and 1.142%. Its gate retains all 1,775 correct geometric poses. On 180 text queries, common GPT assessment gives QGVI-A 90.8–93.9% answer precision at 72.8% coverage. On 250 image-and-instruction queries, common GPT assessment gives QGVI-A 90.4% confirmed correct returns, versus 26.8% for Retrieve + Select and 17.6% for Fixed Search + Verify, at lower precision than the one-shot control. Information and budgets differ. Controls do not establish a consistent retrieval benefit from grid augmentation; photometric refinement worsens median pose errors without improving threshold success. Shared answer accounting retains task-specific correctness: semantic judgments remain model assessments, source-camera recovery is measured separately, and geometric thresholds use source-verified or estimated scales.

1. Introduction

We locate viewpoints within an existing Gaussian asset from text, RGB, or an image with a target-view instruction, using supplied calibrated references. A description may admit several viewpoints; RGB constrains the source camera. An image-and-instruction query can identify a building

while requesting a different framing (Fig. 1); recovering the source camera alone does not establish that the requested view satisfies the instruction.

Several distant poses may satisfy a description. Continuous distributions can represent this ambiguity, but returning satisfying views does not require estimating them. We separate semantic satisfaction from source-camera recovery and retain geometric accuracy for RGB queries.

Gaussian splatting supports fast rendering of reference images at known poses [18]. CityLoC uses Gaussian scenes for text-conditioned pose localization [26]; LoGS combines geometric initialization and render-based refinement for visual localization [8]. Queryable Gaussian View Index (QGVI) provides retrieval and localization tools for this task. QGVI-T retrieves captioned views with stored poses. QGVI-I estimates poses from RGB references and sparse 3D landmarks, then gates them using distinct geometric support. QGVI-A coordinates these tools through a shared record of candidates, poses, and observations. Correct returns, incorrect returns, and coverage share one accounting, with task-specific correctness criteria.

QGVI-T makes the supplied asset searchable through a posed view index. Each entry binds a structured caption to an original image or a rendered observation and its camera pose. BM25 returns candidates after removing duplicate images and poses. Rendering and captioning occur offline; retrieval requires no online Gaussian rendering, VLM call, or scene-specific pose network. Grid rendering is an optional coverage choice. Matched controls with historical fusion do not establish a consistent augmentation benefit; the current BM25 interface's augmentation benefit remains unmeasured.

QGVI-I gates LoGS geometric poses. It counts distinct supported query keypoints, avoiding repeated support across references. A threshold selected on Block A and SF validation queries is frozen across nine test scenes. Accepted poses remain unchanged. An optional photometric ablation reduces false accepts slightly but worsens median pose errors without improving threshold success on this cohort.

QGVI-A's vision-language controller selects these interfaces as needed. It checks supplied predicates, proposes local camera moves or explicit poses, and inspects Gaussian renders. Supported answers exit early; otherwise it searches within shared action and render budgets or returns insufficient evidence. Completed agentic runs use BM25 and the geometric RGB default. The target-view controls compare whole policies with unequal information and budgets; they do not isolate navigation from static verification.

Our contributions are:

- Posed caption retrieval (QGVI-T) and selective geometric localization (QGVI-I), exposing candidate views and geometrically supported poses within a supplied Gaussian asset.
- Adaptive scene querying (QGVI-A), which connects these interfaces through predicate checks, camera proposals, and rendered feedback, with early exit and budgeted abstention.
- A nine-scene evaluation of text, RGB, and image-and-instruction queries, separating semantic satisfaction from source-camera recovery while accounting for answer precision and coverage.

2. Related Work

Text-conditioned localization. Text-conditioned localization connects a description to a position or pose in a scene. Text2Pos and Text2Loc localize within point-cloud maps through coarse-to-fine inference [20, 44]; PMSH models uncertainty when descriptions only partially match submaps [12]. CityLoC [26] instead learns a text-conditioned distribution over camera poses using diffusion and rendered candidates for refinement. Such distributions accommodate multiple plausible poses. Our semantic query task instead asks for a bounded set of satisfying views; source-pose error alone cannot credit every valid alternative. This task distinction does not establish that discrete retrieval universally outperforms continuous estimation.

Semantic scene representations also support spatial queries. CLIP [30] aligns image and text representations, and OpenScene aligns point features with image and text embeddings [28]. LERF, LangSplat, and Feature 3DGS extend language or semantic features to renderable scene representations [19, 29, 47]. QGVI-T searches posed views using BM25 [32]. Our historical fusion variant and matched controls also evaluate MiniLM-based sentence embeddings [31, 43]. Its semantic evaluation measures description satisfaction at returned views; source-camera recovery is evaluated separately.

Geometric localization. Structure-based methods estimate pose from image-to-map correspondences, with retrieval and local matching determining the available geometric evidence. Active Search prioritizes direct 2D–3D matching [36]; hierarchical localization retrieves candidate places before local matching [33]. InLoc further combines dense matching with synthesized-view verification [40].

Global descriptors, including NetVLAD, MixVPR, and SALAD, retrieve reference images [1, 3, 16]. Local representations range from SIFT and its RootSIFT normalization [2, 25] to learned detectors and descriptors such as SuperPoint and D2-Net [10, 11]. SuperGlue and LightGlue assign sparse feature matches [24, 34]; LoFTR establishes detector-free correspondences [39]. For pose recovery, COLMAP supplies sparse landmarks [37], while PnP solvers such as EPnP [21] and robust consensus estimation [13] handle noisy matches. Scene-

coordinate regression learns image-to-scene associations directly: DSAC makes robust estimation differentiable [6], and ACE accelerates scene-specific training [7]. QGVI-I adopts NetVLAD, SuperPoint, SuperGlue, and SfM landmarks through LoGS [8]. Its added acceptance rule counts distinct inlier query keypoints at LoGS's geometric pose, avoiding repeated support across references, and leaves that pose fixed.

Rendering and pose refinement. Pose refinement can use rendered appearance or feature alignment. NeRF represents continuous radiance fields [27]; iNeRF refines an initial pose using rendered-image residuals [45]. Feature-based alternatives include PixLoc's learned multi-scale alignment [35] and MCLoc's pretrained features with particle filtering [42]. 3D Gaussian Splatting supports differentiable novel-view rendering [18]. GSLoc differentiates image-alignment error with coarse-to-fine blurring [5]; LoGS adds affine brightness correction and informative-pixel masks to photometric refinement [8]. GSplatLoc grounds keypoint descriptors in Gaussian maps [38], and SplatHLoc combines rendered and image-derived features at different matching stages [41]. 6DGS estimates pose from image–ray bindings without requiring an initial pose or iterative pose optimization [4]. Rendering optionally supplies QGVI-T's grid views. We evaluate the released LoGS photometric optimizer as an ablation of the geometric default. QGVI-A uses rendered feedback to inspect proposed camera poses against supplied predicates, including source-relative constraints.

Selective prediction. Selective classification separates prediction from withholding an output, exposing a risk–coverage trade-off [15]. For localization, accepting an out-of-map image, rejecting an in-map image, and accepting an inaccurate in-map pose are distinct outcomes. QGVI-I selects its threshold subject to retaining validation-correct poses and holds the underlying estimates fixed in the paired gate ablation. Historical QGVI-T verification withholds unsupported candidates. QGVI-A can search further through camera proposals, terminating early on supported answers or abstaining within a fixed budget. These decisions are evaluated separately from view relevance and pose accuracy.

3. Method

3.1. Task: Querying a Gaussian Scene Asset

The input scene M comprises an existing Gaussian reconstruction and its associated calibrated reference data. The task is to return camera poses within this asset's coordinate frame that satisfy a query. For query q and task type $\tau \in \{\text{text}, \text{source}, \text{target}\}$, define

$$\mathcal{A}_\tau(q, M) = \{T \in SE(3) : C_\tau(q, T; M) = 1\}. \quad (1)$$

Here T is a scene-frame camera pose. For text, C_{text} requires a usable view satisfying the description; several distant poses may qualify. For RGB source localization, C_{source} requires an in-map query and agreement with its source pose within the evaluation thresholds. Out-of-map RGB queries have an empty valid-answer set. For an image with a target-view instruction, C_{target} requires a usable rendered view satisfying every supplied predicate and its relation to the source; recovering the source camera alone is insufficient. These sets define evaluation targets and are not computed exhaustively. QGVI-T retrieves stored poses, QGVI-I estimates a source pose, and QGVI-A coordinates their interfaces with camera exploration. Each may return no answer.

The routes use different reference assets (Fig. 2). QGVI-T indexes original images and optional renders from frozen Horizon-GS reconstructions [17, 18], with captions and camera-to-world poses. QGVI-I uses original training RGB images and sparse landmarks triangulated from feature matches with fixed training cameras [37], using LoGS geometric initialization [8] and a calibrated acceptance rule. Supplementary Sec. B gives implementation settings.

3.2. QGVI-T: Language-Indexed View Retrieval

Posed index construction. The index starts from calibrated reference images. Optional augmentation adds renders from a deterministic lattice near the training cameras, with sampled orientations and training-camera intrinsics. Within each scene and camera domain, views sharing a center share a place identifier. Augmentation adds virtual observations with explicit camera poses; dense pre-rendering is not required by the interface.

Caption retrieval. DeepSeek describes each reference offline using structured fields for objects, attributes, counts, relations, and visible text. The prompt prohibits inferring hidden objects or camera metadata; the uncertainty field is excluded from the index. Reference images and poses remain fixed.

The current caption-search interface uses whole-caption BM25 [32], without embeddings. It ranks positive-score views and removes duplicate pixels or camera poses, returning up to six candidates with their stored cameras. Ranking does not verify semantics or establish that a matching view is absent.

Historical fusion variant. Earlier experiments combine MiniLM [31, 43] and BM25 rankings by reciprocal-rank fusion [9] and return up to five distinct places. These results



Figure 2. **QGVI-A coordinates caption search and source localization.** Call/return links expose tool interfaces; Sec. 4 specifies the evaluated variants. Cached BM25 candidates from the evaluated original-plus-grid bank seed the pictured episode. The agent checks supplied predicates, updates the camera $T = (R, c)$, and renders $\mathcal{R}(M, T, K)$ with fixed intrinsics K. The equations show local camera increments; direct pose proposals are also supported. The RGB example shows a geometric LoGS pose on SfM points. Selected observations omit intermediate proposals; guides mark the left third. Checks are model assessments, coordinates use native units, and images depict synthetic scenes.

retain the fusion label. Supplementary Secs. B and H detail its shortlist and optional visual-verification ablation.

3.3. QGVI-I: Selective Geometric Localization

Geometric initialization. For image I with intrinsics K, the LoGS initializer retrieves ten training images with NetVLAD [3] and matches SuperPoint features [10] using SuperGlue [34]. Reference feature indices connect matches to triangulated landmarks. Correspondences from all references form one PnP problem without covisibility clustering. PnP-RANSAC [13] uses a 12-pixel reprojection threshold and is followed by geometric pose refinement, yielding T_0. Our acceptance rule operates on this geometric estimate without photometric optimization.

Distinct geometric support. Matches from several references can repeat the same query feature. At the geometric camera-to-world pose T_0, we therefore count each query keypoint once, even when it matches several references or competing landmarks. For keypoint u_j, let $\mathcal{P}_j^+(T_0)$ contain its matched landmarks with positive depth. Selecting the smallest reprojection residual gives

$$n_{\mathrm{u}}(T_0) = \sum_j \mathbf{1}\left[\min_{X \in \mathcal{P}_j^+(T_0)} \|\pi_K(T_0^{-1}X) - u_j\|_2 \le 12\right]. \tag{2}$$

The minimum of an empty set is infinity, and π_K is calibrated projection. We reproject the cached geometric correspondences at the saved geometric pose. Support counts distinct query keypoints, not correspondence rows.

Acceptance calibration. We calibrate on 490 Block A and SF validation queries, equally split between in-map and out-of-map examples. A fixed count grid selects the threshold while retaining every correct geometric validation pose separately at 1 m/2° and 5 m/5°. Selection prioritizes out-of-map rejection, then in-map retention, then the smaller threshold. The selected cutoff is 40 distinct inliers. Let $e_{\mathrm{PnP}}(I)$ indicate successful PnP estimation. Frozen across nine test scenes, the rule is

$$a(I) = e_{\mathrm{PnP}}(I)\mathbf{1}[n_{\mathrm{u}}(T_0) \ge 40],$$
$$\widehat{T}(I) = \begin{cases} T_0, & a(I) = 1, \\ \varnothing, & a(I) = 0. \end{cases} \tag{3}$$

The gate preserves T_0. The optional "QGVI-I + photo." ablation runs up to 20 native LoGS photometric steps with brightness correction and informative-pixel masks, then gates the refined pose $T_\star$ under native LoGS eligibility. Its independently calibrated distinct-count cutoff is also 40. Neither gate guarantees correct-pose retention on unseen queries.

3.4. QGVI-A: Adaptive Scene Querying

Shared evidence and constraints. QGVI-A calls QGVI-T's caption-search and QGVI-I's source-localization interfaces (Fig. 2) using shared candidates, camera poses, observations, geometric support, and unmet task-supplied predicates. RGB calls use the source image; text calls retrieve candidate cameras. The controller inspects target renders and records each predicate as satisfied, violated, or unknown, citing observations. Acceptance requires every predicate and the source relation to be supported by these model assessments.

Exploration and stopping. An ordinary retrieval/localization prefix first attempts a direct answer: a source pose must pass the geometric gate; a target view must pass the rendered-predicate check. Otherwise, its evidence enters adaptive search. For camera-to-world pose $T = (R, c)$, local actions update $c' = c + R\Delta t$ and $R' = R\Delta R$; the controller may also propose a bounded valid $SE(3)$ pose. Rendering $I' = \mathcal{R}(M, T', K)$ with fixed intrinsics supplies the next observation. Inspection and camera updates continue within shared decision, render, and time budgets. At termination, the host returns a saved supported answer or insufficient evidence. Sec. 4 and Supplementary Sec. E specify evaluated tool variants and budgets.

4. Experiments

We evaluate text retrieval, RGB localization, target-view search, rejection, and latency across nine scenes; supplementary sections give protocol details.

4.1. Experimental Setup

Datasets and reference views. The nine-scene benchmark combines Block Small and Block A from MatrixCity [22], NYC and SF from UC-GS [46], and Citysample, Colosseum, Elvenruin, Park, and Road from the Horizon-GS dataset [17]. Seven scenes are synthetic; Park and Road contain real captures. We use frozen Horizon-GS reconstructions across all nine scenes to generate reference views and novel-view queries. The BM25 bank contains 18,570 original-image entries and 30,522 grid renders (49,092 entries); rendered-bank controls use the latter. The source split has no exact file overlap, but some evaluation observations contributed to reconstruction.

Query cohorts. The main text cohort contains 180 DeepSeek-authored descriptions, 20 per scene: 74 from original test images and 106 from novel Gaussian renders; six are unsupported by their source images. Earlier 176-query and 135-query cohorts remain separate.

RGB testing uses 1,839 in-map and 1,839 out-of-map queries. Positives comprise 1,029 held-out source images, including synthetic imagery, and 810 novel Gaussian renders. Negatives comprise 1,029 training images and 810 renders from disjoint scene worlds within the nine scenes. Query composition varies by scene; Block A supplies 32.7% of positives. Methods share frozen queries, splits, calibration, and target scenes. Four scene scales are source-verified and five DA3-estimated [23] (Supplementary Table S11); calibration and interpretation are specified in Supplementary Sec. C.

Compared methods. Our CityLoC [26] reimplementation uses three fixed seeds and 64 pose proposals. QGVI-T returns the first BM25 candidate from the original-plus-grid bank; QGVI-A returns its saved final answer. Historical fusion and verification controls are retained in the supplement. QGVI-I uses the LoGS [8] geometric initializer and accepts successful PnP poses with at least 40 distinct inlier query keypoints. The optional ablation adds native photometric refinement ("+ photo."). QGVI-I shares LoGS's native assets; SplatHLoc [41] also uses its native map. Native LoGS accepts retrieval fallbacks; SplatHLoc requires four final inliers. We calibrate count and ratio gates for both LoGS pose variants, and a count gate for SplatHLoc on the same 490 Block A/SF validation queries, preserving correct poses, then freeze thresholds across nine scenes.

Model judgments. The main text comparison uses one GPT-6.1 Sol low-effort relevance/validity rubric for all frozen first answers. Compatible historical judgments are reused; new reviews preserve query text and image bytes. Request transports differ, as disclosed in Supplementary Sec. E. DeepSeek supplies captions and descriptions. Earlier Qwen/DeepSeek and Kimi/GPT assessments remain separate (Supplementary Sec. D.2). Matched retrieval controls retain their own judging protocol.

Agentic evaluation. QGVI-A uses early exit, bounded search, BM25 over the original-plus-grid bank, and geometric QGVI-I. Alongside text/RGB, we test 250 image-and-instruction queries. *Retrieve + Select* makes one choice among the same six initial BM25 renders. *Fixed Search + Verify* checks six precommitted cameras around the first candidate, stopping at the first supported view. All use GPT-6.1 Sol low and at most six render attempts, with separate high-effort predicate assessments. Information access and

Table 1. **Shared answer accounting and net query utility.** First-answer results pooled over nine scenes. N counts query–seed observations for CityLoC and queries otherwise. Net utility is $100(S - W)/N$. Semantic correctness requires a usable view satisfying the description or all target predicates. RGB correctness requires an in-map pose within 1 m/2°; five scenes have DA3-estimated scales. Ranges propagate uncertain GPT judgments, not sampling uncertainty. Panel (a) uses the common GPT-low relevance/validity rubric. Panels use different cohorts or judging protocols and must not be ranked against each other. Semantic panels bold best conservative endpoints; RGB bold marks best values, including ties. Coverage is descriptive.

Method	N	Answered C (%)	Answer precision P (%) ↑	Correct returns Y (%) ↑	Wrong returns E (%) ↓	Net utility $100U_1$ ↑
(a) Text: common GPT-low relevance and validity judgments						
CityLoC w/o refinement	540	100.00	26.67–66.48	26.67–66.48	33.52–73.33	-46.67–32.96
CityLoC with refinement	540	80.19	36.26–61.66	29.07–49.44	30.74–51.11	-22.04–18.70
QGVI-T	180	100.00	60.00–70.56	60.00–70.56	29.44–40.00	20.00–41.11
QGVI-A	180	72.78	**90.84–93.89**	**66.11–68.33**	**4.44–6.67**	**59.44–63.89**
(b) RGB: geometric correctness, 50% out-of-map						
LoGS, native	3,678	100.00	48.26	**48.26**	51.74	-3.48
SplatHLoc, calibrated count	3,678	99.54	30.73	30.59	68.95	-38.36
QGVI-I	3,678	49.78	96.94	**48.26**	1.52	46.74
QGVI-I + photo.	3,678	49.70	**97.10**	**48.26**	**1.44**	**46.82**
QGVI-A	3,678	49.78	96.83	48.21	1.58	46.63
(c) Image-and-instruction target views: GPT high-effort judgments						
Retrieve + Select	250	27.20	**98.53**	26.80	**0.40**	26.40
Fixed Search + Verify	250	20.00	88.00–90.00	17.60–18.00	2.00–2.40	15.20–16.00
QGVI-A	250	97.60	92.62–93.03	**90.40–90.80**	6.80–7.20	**83.20–84.00**

call budgets differ (Supplementary Secs. E and F); text retains its common low-effort rubric.

Common answer metrics. Let A count first returns among N queries, with S correct and W wrong. Coverage is $C = A/N$, answer precision $P = S/A$, correct-return rate $Y = S/N$, and wrong-return rate $E = W/N$. Uncertain judgments bound S and W without changing A. Net query utility $U_\lambda = Y - \lambda E$ assigns reward $+1$ to a correct return, $-\lambda$ to a wrong return, and zero to no return [14]. Table 1 uses $U_1 = Y - E$; equal reward and penalty are illustrative, not calibrated deployment costs. Correctness follows the panel-specific criteria in the caption. Utility does not replace pose error or match acceptance policies; Supplementary Sec. E gives cost sensitivity.

Text metrics. Top-K success (Hit@K) counts queries with a valid image satisfying all textual constraints among the first K outputs; empty lists fail. The 176-query cohort measures relevance only. Lower bounds count definite successes; upper bounds also count uncertain outcomes. These are label bounds, not confidence intervals or full-corpus recall. No-result rate counts empty returns, excluding runtime failures and missing judgments; it cannot certify that no matching view exists. Scene-macro scores weight scenes equally after CityLoC seed averaging; pooled scores weight queries.

RGB metrics. An in-map query depicts the target scene; an out-of-map query depicts another scene world. Translation error is camera-center distance; rotation error is the relative rotation angle. Pose success divides accepted poses within both 1 m/2° (strict) or 5 m/5° (loose) by *all in-map queries*, including rejected or missing poses as failures. Metre thresholds include five estimated scales. With in-map indicator y and acceptance indicator a,

$$\mathrm{FAR} = \frac{\sum(1-y)a}{\sum(1-y)}, \qquad \mathrm{FRR} = \frac{\sum y(1-a)}{\sum y}. \tag{4}$$

FAR and FRR use all out-of-map and in-map queries, respectively, and are independent of metre scale. Accepted wrong in-map poses are localization failures. Accepted-pose precision divides correct poses by all accepted outputs; conditional in-map precision excludes out-of-map outputs. Precision depends on the query mixture, balanced here.

4.2. Answer Quality and Localization Results

First-answer utility. Table 1 compares text methods under one judging rubric. QGVI-A yields 59.44–63.89 net reward units per 100 text queries; BM25 QGVI-T yields 20.00–41.11. On target-view queries, QGVI-A yields 83.20–84.00, versus 26.40 for Retrieve + Select and 15.20–16.00 for Fixed Search + Verify. Confirmed correct-return rates are 90.4%, 26.8%, and 17.6%, respectively. The one-shot control has higher precision (98.53%) at 27.2% coverage; QGVI-A answers 97.6% at 92.62–93.03% precision.

This compares complete policies, not navigation alone. Supplementary Sec. F gives scene, task-family, and paired uncertainty analyses. On RGB, QGVI-A yields 46.63, below QGVI-I's 46.74 and its photometric ablation's 46.82. Its 1,831 returns comprise 1,773 strict successes, 46 inaccurate in-map poses, and 12 out-of-map accepts.

Text-view retrieval. BM25 QGVI-T returns 180/180 answers, with 108 confirmed correct, 53 incorrect, and 19 uncertain. QGVI-A returns 131/180, with 119 confirmed correct, 8 incorrect, and 4 uncertain. Supplementary Table S8 includes fusion and verification under the same rubric; Table S9 preserves earlier judgments. All assessments use frozen localization outputs.

Source poses and semantic judgments. On these 180 descriptions, QGVI-T fusion achieves 0.6% Top-1 source-pose success and 3.3% Pose Recall@5 at 5 m/5°; full-bank oracle coverage is 22.2%. Semantic success does not establish source-camera accuracy (Supplementary Sec. D).

RGB localization and acceptance. QGVI-I achieves 97.3% and 97.4% scene-macro pose success at 1 m/2° and 5 m/5° (Supplementary Table S11(b)). It accepts 13/1,839 out-of-map queries (0.707% pooled; 0.764% scene mean) and rejects 21/1,839 in-map queries (1.142% pooled; 0.765% scene mean). Accepted in-map poses include 43 strict and 40 loose localization failures. LoGS and SplatHLoc achieve 97.3% and 76.0% strict success, with 100.0% and 99.8% scene-mean false acceptance under native output rules. These native rates do not measure the benefit over calibrated gates. Every NYC and SF positive lies within 0.59 m of a training camera, with shared reconstruction dependencies; difficulty varies across scenes.

The QGVI-I gate preserves its 1,831 retained geometric pose matrices (Table S19) and all 1,775 strict and 1,778 loose successes. The photometric ablation retains these success counts and rejects three more distractors; precision rises from 96.94% to 97.10%. Under matched validation-retention objectives (Table 2), raw and distinct initializer counts accept 24 and 13 distractors, respectively; the photometric ablation's final distinct support accepts ten. All retain the same strict-success queries, with different rejection and computational costs.

Photometric refinement changes all 3,678 pose matrices without strict or loose success/failure transitions. In-map median translation/rotation errors increase from 0.012 m/0.023° to 0.028 m/0.085°. Translation and rotation errors worsen for 78.0% and 93.0% of in-map queries. We therefore retain the geometric estimate by default (Supplementary Sec. K).

Table 2. **Rejection under a shared calibration objective.** Thresholds retain every validation-correct pose, then maximize rejection of validation distractors. Pooled test rates (%) use 1,839 queries per class; pose success is at 1 m/2°. QGVI-I gates the geometric initializer; "+ photo." adds native LoGS refinement and the final-pose gate. SplatHLoc retains its native requirement of at least four inliers. Bold marks the best displayed rates and ties.

Method / support threshold	Pose ↑	FAR ↓	FRR ↓
Initializer: count ≥ 40	**96.52**	1.305	1.033
Initializer: ratio ≥ 0.1	96.36	79.445	0.761
QGVI-I: distinct ≥ 40	**96.52**	0.707	1.142
LoGS: pairs ≥ 40	**96.52**	1.251	1.142
QGVI-I + photo.: ≥ 40	**96.52**	**0.544**	1.142
SplatHLoc: native (≥ 4)	61.17	99.347	**0.272**

Table 3. **Matched text retrieval controls.** Same 180 queries, 30,522 rendered candidates, place selection, and five-result budget; no online verification. GPT judgments are complete. Ranges are label-uncertainty bounds (%), not confidence intervals. Bold marks the largest lower bound; overlapping ranges do not establish a ranking.

Retrieval	Top-1 ↑	Top-5 ↑
CLIP	39.4–54.4	62.2–80.6
BM25	**62.8**–78.9	**77.2**–92.2
MiniLM	56.7–73.9	70.6–90.0
QGVI-T (fusion)	61.1–77.8	73.9–91.7

4.3. Matched Text Retrieval Controls

Table 3 compares CLIP, BM25, MiniLM, and historical fusion on the same 180 descriptions and rendered bank: 64 candidates, a 20-view shortlist, five distinct places, and no online verification. All 2,825 output pairs have GPT judgments. BM25's confirmed Top-1/Top-5 (62.8%/77.2%) exceeds fusion's (61.1%/73.9%); overlapping label bounds do not establish a fusion benefit. Supplementary Sec. I gives per-scene results and partial Kimi assessments.

Optional verification remains a separate operating-point ablation. Under the historical Qwen/DeepSeek rubric it raises Top-1 from 58.3% to 66.1% but lowers Top-5 from 78.3% to 72.8%, with 45/180 empty returns. The earlier 176-query controls and query-type breakdown are retained in Supplementary Tables S20 and S21.

4.4. Contribution of Rendered Grid Views

We add 30,522 Gaussian grid renders to 18,570 original images, fixing captions, historical fusion, queries, and the five-result budget. BM25 statistics are recomputed per bank. Original entries are judged using Gaussian renders at their stored cameras. All 1,329 output pairs have GPT judgments; original-only predictions are reused unchanged.

Table 4 shows four additional confirmed Top-1 and two Top-5 successes across 180 descriptions. Gains occur

Table 4. **Adding grid views to original images.** Historical fusion compares 18,570 original training images (O) with O plus 30,522 Gaussian grid renders (O+G). Each query searches its own scene. GPT-confirmed success (%); uncertain labels remain in the denominator. Bold marks higher confirmed success within each row and metric, including ties.

Description source	n	Top-1 ↑		Top-5 ↑	
		O	O+G	O	O+G
All descriptions	180	57.8	**60.0**	73.3	**74.4**
Original-image source	74	**52.7**	**52.7**	**71.6**	70.3
Novel-render source	106	61.3	**65.1**	74.5	**77.4**

Table 5. **Measured query latency.** Median/nearest-rank 95th percentile in seconds, pooled over all nine scenes. Text: 180 queries (540 observations per CityLoC arm); RGB: 36 fixed queries, three repeats. Empty outputs remain in the denominators. QGVI-T (fusion) denotes historical CPU retrieval without online VLM calls. Text timer boundaries differ; bold marks the fastest text latency and compares qualified RGB latencies.

Method	Query time (s) ↓ Median / p95
CityLoC w/o refinement	3.376 / 8.389
CityLoC with refinement	171.720 / 429.821
QGVI-T (fusion)	**0.020 / 0.063**
LoGS	0.436 / 2.293
SplatHLoc	0.510 / 0.847
QGVI-I	**0.240 / 0.450**

among novel-render descriptions; the 74 original-image descriptions have no Top-1 gain and one fewer Top-5 success. All paired label-uncertainty bounds include zero, and source groups differ in scene proportions. Thus, the 2.64-fold bank expansion has no established consistent benefit; it also does not test the current BM25 interface's augmentation gain. Supplementary Sec. J reports same-pose, grid-only, density, and fixed-size replacement controls.

4.5. Query Latency

QGVI-T fusion takes 0.020/0.063 s median/p95 including CPU encoding, retrieval, and place selection (Table 5). Text timers and hardware differ; ranking-only CLIP/BM25/MiniLM timings are excluded.

RGB timing uses four queries per scene, three resident repeats, and an RTX PRO 6000 Blackwell GPU. Timers include decoding, preprocessing, localization, acceptance, and device synchronization, excluding map/model loading. QGVI-I takes 0.240/0.450 s median/p95. Its photometric ablation, including the final gate, has not been timed; native LoGS timing excludes that gate. Supplementary Sec. D.3 gives per-scene times, timer boundaries, and repeatability checks.

5. Discussion and Limitations

Refinement and acceptance. Correct-pose retention is empirical: accurate poses below 40 distinct inliers may be rejected. Geometric support provides no distribution-free risk bound; the calibrated distinct-count operating point does not minimize integrated risk across thresholds.

Text success and abstention. Historical fusion verification's 45/180 empty returns and QGVI-A's 49/180 nonreturns do not prove absence of a satisfying view. Rejecting cross-scene RGB distractors does not establish robustness to arbitrary appearance, sensor, or content changes.

Retrieval and view coverage. Optional grid expansion enlarges the bank 2.64-fold without an established consistent benefit. BM25 and agentic runs retain this bank. Label bounds are not statistical significance tests.

Evaluation scope. Camera moves need not be traversable; predicate checks cannot establish scene-wide answerability. QGVI-A has more correct target-view returns but lower answer precision than Retrieve + Select (92.6–93.0% versus 98.5%). Information, call budgets, and transports differ, precluding a controlled latency ranking. Source-preserving static verification with matched information and call budgets remains untested. Querying a Gaussian asset does not establish superiority over other scene representations. Model-judge agreement measures consistency rather than human correctness. DA3 supplies a fixed scale conversion where source metric scales are unavailable, not surveyed ground truth. Reconstruction reuse and scale calibration are specified in Supplementary Sec. C.

6. Conclusion

Queryable Gaussian View Index connects caption retrieval, selective geometric localization, and camera exploration within supplied Gaussian assets. QGVI-A reduces wrong text returns relative to BM25 while answering fewer queries. On 250 target-view queries, GPT judges 226 answers correct, versus 67 for Retrieve + Select and 44 for Fixed Search + Verify. QGVI-A returns more wrong answers; information and budgets differ. QGVI-I retains every correct geometric pose on the RGB cohort while rejecting most distractors. Controls support a geometric default; neither grid augmentation nor photometric refinement has an established accuracy advantage here. Evaluation retains task-specific correctness and model-judgment limitations.

References

[1] Amar Ali-bey, Brahim Chaib-draa, and Philippe Giguère. MixVPR: Feature mixing for visual place recognition. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 2998–3007, 2023.

[2] Relja Arandjelović and Andrew Zisserman. Three things everyone should know to improve object retrieval. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2911–2918, 2012.

[3] Relja Arandjelovic, Petr Gronat, Akihiko Torii, Tomas Pajdla, and Josef Sivic. NetVLAD: CNN architecture for weakly supervised place recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5297–5307, 2016.

[4] Matteo Bortolon, Theodore Tsesmelis, Stuart James, Fabio Poiesi, and Alessio Del Bue. 6DGS: 6D pose estimation from a single image and a 3D gaussian splatting model. In *European Conference on Computer Vision (ECCV)*, pages 420–436, 2024.

[5] Kazii Botashev, Vladislav Pyatov, Gonzalo Ferrer, and Stamatios Lefkimmiatis. GSLoc: Visual localization with 3d gaussian splatting. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 5664–5671, 2024.

[6] Eric Brachmann, Alexander Krull, Sebastian Nowozin, Jamie Shotton, Frank Michel, Stefan Gumhold, and Carsten Rother. DSAC - differentiable RANSAC for camera localization. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6684–6692, 2017.

[7] Eric Brachmann, Tommaso Cavallari, and Victor Adrian Prisacariu. Accelerated coordinate encoding: Learning to relocalize in minutes using RGB and poses. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5044–5053, 2023.

[8] Yuzhou Cheng, Jianhao Jiao, Yue Wang, and Dimitrios Kanoulas. LoGS: Visual localization via gaussian splatting with fewer training images. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 15029–15036, 2025.

[9] Gordon V. Cormack, Charles L. A. Clarke, and Stefan Büttcher. Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759, 2009.

[10] Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. SuperPoint: Self-supervised interest point detection and description. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 224–236, 2018.

[11] Mihai Dusmanu, Ignacio Rocco, Tomas Pajdla, Marc Pollefeys, Josef Sivic, Akihiko Torii, and Torsten Sattler. D2-Net: A trainable CNN for joint description and detection of local features. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8092–8101, 2019.

[12] Mingtao Feng, Longlong Mei, Zijie Wu, Jianqiao Luo, Fenghao Tian, Jie Feng, Weisheng Dong, and Yaonan Wang. Partially matching submap helps: Uncertainty modeling and propagation for text to point cloud localization. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 8296–8305, 2025.

[13] M. A. Fischler and R. C. Bolles. Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography. *Communications of the ACM*, 24(6):381–395, 1981.

[14] Vojtech Franc, Daniel Prusa, and Vaclav Voracek. Optimal strategies for reject option classifiers. *Journal of Machine Learning Research*, 24(11):1–49, 2023.

[15] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, 2017.

[16] Sergio Izquierdo and Javier Civera. Optimal transport aggregation for visual place recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 17658–17668, 2024.

[17] Lihan Jiang, Kerui Ren, Mulin Yu, Linning Xu, Junting Dong, Tao Lu, Feng Zhao, Dahua Lin, and Bo Dai. Horizon-GS: Unified 3D gaussian splatting for large-scale aerial-to-ground scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 26789–26799, 2025.

[18] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3D gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42 (4), 2023.

[19] Justin Kerr, Chung Min Kim, Ken Goldberg, Angjoo Kanazawa, and Matthew Tancik. LERF: Language embedded radiance fields. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 19729–19739, 2023.

[20] Manuel Kolmet, Qunjie Zhou, Aljoša Ošep, and Laura Leal-Taixé. Text2Pos: Text-to-point-cloud cross-modal localization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6687–6696, 2022.

[21] Vincent Lepetit, Francesc Moreno-Noguer, and Pascal Fua. EPnP: An accurate O(n) solution to the PnP problem. *International Journal of Computer Vision*, 81(2):155–166, 2009.

[22] Yixuan Li, Lihan Jiang, Linning Xu, Yuanbo Xiangli, Zhenzhi Wang, Dahua Lin, and Bo Dai. MatrixCity: A large-scale city dataset for city-scale neural rendering and beyond. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 3205–3215, 2023.

[23] Haotong Lin, Sili Chen, Jun Hao Liew, Donny Y. Chen, Zhenyu Li, Guang Shi, Jiashi Feng, and Bingyi Kang. Depth Anything 3: Recovering the visual space from any views. arXiv preprint arXiv:2511.10647, 2025.

[24] Philipp Lindenberger, Paul-Edouard Sarlin, and Marc Pollefeys. LightGlue: Local feature matching at light speed. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 17627–17638, 2023.

[25] David G. Lowe. Distinctive image features from scale-invariant keypoints. *International Journal of Computer Vi-*

sion, 60(2):91–110, 2004.

[26] Qi Ma, Runyi Yang, Bin Ren, Nicu Sebe, Ender Konukoglu, Luc Van Gool, and Danda Pani Paudel. CityLoc: 6dof pose distributional localization for text descriptions in large-scale scenes with gaussian representation. arXiv preprint arXiv:2501.08982, 2025.

[27] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. NeRF: Representing scenes as neural radiance fields for view synthesis. In *European Conference on Computer Vision (ECCV)*, pages 405–421, 2020.

[28] Songyou Peng, Kyle Genova, Chiyu "Max" Jiang, Andrea Tagliasacchi, Marc Pollefeys, and Thomas Funkhouser. OpenScene: 3D scene understanding with open vocabularies. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 815–824, 2023.

[29] Minghan Qin, Wanhua Li, Jiawei Zhou, Haoqian Wang, and Hanspeter Pfister. LangSplat: 3D language gaussian splatting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 20051–20060, 2024.

[30] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, pages 8748–8763, 2021.

[31] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992, 2019.

[32] Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009.

[33] Paul-Edouard Sarlin, Cesar Cadena, Roland Siegwart, and Marcin Dymczyk. From coarse to fine: Robust hierarchical localization at large scale. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12716–12725, 2019.

[34] Paul-Edouard Sarlin, Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. SuperGlue: Learning feature matching with graph neural networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4938–4947, 2020.

[35] Paul-Edouard Sarlin, Ajaykumar Unagar, Måns Larsson, Hugo Germain, Carl Toft, Viktor Larsson, Marc Pollefeys, Vincent Lepetit, Lars Hammarstrand, Fredrik Kahl, and Torsten Sattler. Back to the feature: Learning robust camera localization from pixels to pose. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3247–3257, 2021.

[36] Torsten Sattler, Bastian Leibe, and Leif Kobbelt. Efficient & effective prioritized matching for large-scale image-based localization. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 39(9):1744–1756, 2017.

[37] Johannes L. Schönberger and Jan-Michael Frahm. Structure-from-motion revisited. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4104–4113, 2016.

[38] Gennady Sidorov, Malik Mohrat, Denis Gridusov, Ruslan Rakhimov, and Sergey Kolyubin. GSplatLoc: Grounding keypoint descriptors into 3d gaussian splatting for improved visual localization. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 12601–12607, 2025.

[39] Jiaming Sun, Zehong Shen, Yuang Wang, Hujun Bao, and Xiaowei Zhou. LoFTR: Detector-free local feature matching with transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8922–8931, 2021.

[40] Hajime Taira, Masatoshi Okutomi, Torsten Sattler, Mircea Cimpoi, Marc Pollefeys, Josef Sivic, Tomas Pajdla, and Akihiko Torii. InLoc: Indoor visual localization with dense matching and view synthesis. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7199–7209, 2018.

[41] Huaqi Tao, Bingxi Liu, Guangcheng Chen, Fulin Tang, Li He, and Hong Zhang. Hierarchical visual relocalization with nearest view synthesis from feature gaussian splatting. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 40981–40991, 2026.

[42] Gabriele Trivigno, Carlo Masone, Barbara Caputo, and Torsten Sattler. The unreasonable effectiveness of pre-trained features for camera pose refinement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12786–12798, 2024.

[43] Wenhui Wang, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. In *Advances in Neural Information Processing Systems*, 2020.

[44] Yan Xia, Letian Shi, Zifeng Ding, João F. Henriques, and Daniel Cremers. Text2Loc: 3D point cloud localization from natural language. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14958–14967, 2024.

[45] Lin Yen-Chen, Pete Florence, Jonathan T. Barron, Alberto Rodriguez, Phillip Isola, and Tsung-Yi Lin. iNeRF: Inverting neural radiance fields for pose estimation. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 1323–1330, 2021.

[46] Saining Zhang, Baijun Ye, Xiaoxue Chen, Yuantao Chen, Zongzheng Zhang, Cheng Peng, Yongliang Shi, and Hao Zhao. Drone-assisted road gaussian splatting with cross-view uncertainty. In *British Machine Vision Conference (BMVC)*, 2024.

[47] Shijie Zhou, Haoran Chang, Sicheng Jiang, Zhiwen Fan, Zehao Zhu, Dejia Xu, Pradyumna Chari, Suya You, Zhangyang Wang, and Achuta Kadambi. Feature 3DGS: Supercharging 3D gaussian splatting to enable distilled feature fields. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 21676–21685, 2024.

A. Supplementary Material

A.1. Historical Gaussian-Depth Pipeline

The historical Gaussian-depth pipeline (v3) localizes images using Gaussian-rendered RGB-D references and geometric verification. It differs from QGVI-I in retrieval, local matching, 3D geometry, and verification. Table S1 compares these complete designs; Table S19 isolates the QGVI-I acceptance gate at fixed geometric initializer poses.

The historical pipeline uses SALAD [16] retrieval with four diverse starting views, RootSIFT [2] correspondences based on SIFT [25], geometric pose estimation, and verification against a newly rendered Gaussian RGB-D view. The query is resized to a maximum side of 512 pixels before SALAD's standard 322-pixel input transform. The reference bank, 512-pixel local features, solver, gates, and render budgets remain fixed. CLIP embeddings are retained for diagnostics and do not determine geometric acceptance. This variant uses the original Horizon-GS [17] checkpoints and frozen native reference banks.

Each reference is back-projected with its own intrinsics. Pose estimation and verification renders use the query intrinsics, so retrieval does not require the reference and query cameras to have matching calibration.

The historical pipeline achieves 81.1% strict scene-macro pose success, accepts no out-of-map queries, and rejects 381 in-map queries (20.7% pooled; 16.0% scene mean). It also accepts 45 inaccurate poses; these localization failures are distinct from false rejects. QGVI-I has 97.3% strict scene-macro pose success, 0.707% pooled false acceptance, and 1.142% pooled false rejection on the same queries. The historical pipeline rejects more in-map queries while accepting no out-of-map query in this cohort. Neither result establishes a universal rejection guarantee.

A.2. Representation and Initialization Diagnostics

Map representation. With the original maps, the historical pipeline accepts 1,458 in-map poses, of which 1,413 satisfy 1 m/2°. Pose success is 76.8% when pooled over all 1,839 in-map queries and 81.1% when scenes receive equal weight; neither denominator excludes rejections. On identical query images and reference camera poses, the earlier common-SH3 adaptation achieves 19/139 strict successes in Block Small and 9/602 in Block A, compared with 88/139 and 365/602 for the original-map system. Reference renders, depth, retrieval encoder and selections, and camera adapter change together, preventing attribution to Gaussian map quality alone. The original maps follow the benchmark configuration, but the transition was evaluated after inspecting earlier outcomes. This is therefore a development comparison; the common-SH3 scores remain separate from Table S1.

Query composition. The earlier 96.0% result uses a separate set of 200 rendered Block A queries, equally split between aerial and street views, inside a supported 256-patch atlas with 24,576 references. That cohort is distinct from the 602 Block A positives in the nine-scene benchmark. Using the same atlas on these 602 queries yields 97 strict successes (16.1%), compared with 365 (60.6%) for the historical SALAD-initialized system using the original representative bank. This comparison changes the initializer as well as reference coverage. The 96.0% estimate therefore characterizes the supported-atlas setting; it does not describe performance on the nine-scene pool.

A source-image comparison identifies 374 queries shared with the earlier Block A benchmark. Source pixels, camera rotations, metre-converted translations and resized intrinsics agree for every matched image. The earlier CLIP-initialized native-v3 run has 113 strict successes (30.2%); the historical SALAD-initialized system has 231 (61.8%), versus 306 (81.8%) for the archived Gaussian-only SALAD control. Cohort membership and pose labels alone therefore do not explain the remaining gap. The control changes retrieval, reference views, local feature resolution to 960 pixels, and verification; it is excluded from the main method comparison.

Camera calibration. An earlier same-intrinsics restriction excluded 16 of Park's 228 in-map queries and limited some Road queries to 12 of 2,669 references. Removing that restriction and rendering at query calibration increases strict success from 80 to 109 queries in Park and from 47 to 75 in Road. Colosseum and Elvenruin each lose one success; the other scenes are unchanged. The restricted adapter rejected 1,518 negatives without matching, whereas the corrected CLIP-initialized system matches all 1,839. With CLIP, Block A retains 172 strict successes; of its 2,408 positive-query initial hypotheses, 1,324 have insufficient matches and 590 fail RANSAC. These counts locate the dominant loss at initialization; they do not isolate retrieval from appearance or reference coverage.

Initializer selection. A paired comparison on all 988 validation queries holds the reference bank, local geometry, budget of four starting views, and acceptance gates fixed. Replacing CLIP ranking with SALAD improves strict scene-macro pose success from 62.6% to 80.0%, with zero accepts among 494 negatives in each arm. The selection rule requires higher strict success, no increase in either scene-macro false acceptance or total negative accepts, and no additional execution failures. The SALAD recipe was fixed before the full test run and was not retuned on its outcomes. On the complete test cohort, strict success increases from 62.7% to 81.1%, including 172 to 365 strict successes in Block A. Both query pools retain historical development

Table S1. **Historical Gaussian-depth pipeline and QGVI-I.** Both use the same nine-scene development queries and fixed scale factors. Pose success uses all in-map queries; false acceptance uses all out-of-map queries; false rejection uses all in-map queries. Rates are percentages; scene means weight scenes equally. † DA3-estimated scale. This comparison changes several components jointly and does not isolate the acceptance gate. Bold marks best values within each scene and metric, including ties.

Method	Metric (%)	Block Small	Block A	NYC	SF	Citysample	Colosseum	Elvenruin	Park	Road	Scene mean
In-map queries	n	139	602	180	174	142	126	118	228	130	—
Out-of-map queries	n	139	602	180	174	142	126	118	228	130	—
Scale	m / native unit	25	25	1	1	24.777†	12.287†	14.135†	1.040†	1.339†	—
Gaussian-depth pipeline (v3)	Pose 1 m/2° ↑	63.3	60.6	**100.0**	**100.0**	88.0	64.3	94.9	82.9	76.2	81.1
	Pose 5 m/5° ↑	64.0	62.0	**100.0**	**100.0**	95.8	64.3	97.5	84.2	78.5	82.9
	False accept ↓	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
	False reject ↓	36.0	37.7	**0.0**	**0.0**	4.2	29.4	2.5	14.0	20.0	16.0
QGVI-I	Pose 1 m/2° ↑	**90.6**	**93.4**	**100.0**	**100.0**	**100.0**	**91.3**	**100.0**	**100.0**	**100.0**	**97.3**
	Pose 5 m/5° ↑	**92.1**	**93.5**	**100.0**	**100.0**	**100.0**	**91.3**	**100.0**	**100.0**	**100.0**	**97.4**
	False accept ↓	**0.000**	0.332	1.111	4.023	1.408	**0.000**	**0.000**	**0.000**	**0.000**	0.764
	False reject ↓	**3.597**	**2.492**	**0.000**	**0.000**	**0.000**	**0.794**	**0.000**	**0.000**	**0.000**	**0.765**

exposure.

Unresolved geometric conventions. The source audit identifies an unexpressed half-pixel conversion between Gaussian raster coordinates and OpenCV feature-depth lifting. A conditional CPU consistency replay changes initial poses without changing gates; the source-image pixel convention and final localization effect remain unresolved. No calibration correction is applied to the historical pipeline in Table S1. Repeated SIFT orientations can also count one physical landmark more than once in correspondence support. These limitations remain separate from the measured initializer comparison.

A.3. Reference-View Coverage

Orientation coverage is evaluated in the historical CLIP-initialized common-SH3 adaptation using three reference banks: the baseline grid, an equal-size bank replacing selected views with upward-facing views at the same grid positions, and a bank containing all additional upward-facing views. The equal-size comparison holds the reference count fixed and retains every query. On Block Small, both modified banks add three successful queries at each pose threshold: success at 1 m/2° increases from 19/139 (13.7%) to 22/139 (15.8%). Full expansion adds no further successes over equal-size replacement. Block A remains at 9/602 (1.5%) for both thresholds under all three banks. None of the six runs accepts an out-of-map distractor. These results concern the two exposed development scenes and the selected distractor pools.

A.4. Historical Retrieval Measurements

For the historical pipeline, retrieval timing includes reading, decoding, and resizing the query, SALAD and retained CLIP encoding, and SALAD ranking with selection of four starting views. GPU synchronization follows encoding. Model/index loading, pose search, render verification, and output writing are excluded; diagnostic descriptor writes between timed stages are also excluded. These measurements characterize the historical initialization stages, not optimized production latency or QGVI-I. Table S3 also preserves CLIP-based timings from separate camera-support configurations and the 200-query Block A atlas. Their cohorts and configurations remain distinct.

B. Implementation Details

Posed view bank. The historical bank has 30,522 renders at 2,330 places. Centers form a deterministic lattice near training cameras, borrowing their intrinsics and interpolating nearby camera heights. A place identifier combines scene, camera domain, and center hash, grouping orientations at one center; it is neither a semantic landmark nor a radius cluster. Original-only and original-plus-grid controls isolate augmentation (Sec. 4.4); Supplementary Sec. J reports density and replacement.

QGVI-T caption search. Retrieval indexes caption summaries, objects, attributes, relations, and visible text, excluding uncertainty. BM25 uses $k_1 = 1.2$, $b = 0.75$. The current interface adds 18,570 originals to the rendered bank. It returns up to six positive-score candidates, deduplicating image hashes and full poses rounded to six decimals; view IDs break ties. No embeddings are loaded.

Historical fusion and verification. Earlier comparisons pool maximum MiniLM sentence cosine similarity per view [31, 43] and fuse semantic and BM25 ranks [9]: $s_i = (60 + r_i^{\text{sem}})^{-1} + \mathbf{1}[b_i > 0](60 + r_i^{\text{lex}})^{-1}$, with one-based ranks and BM25 score b_i. View IDs break ties. Among

Table S2. **Auxiliary common-map reference-bank ablation for RGB localization.** These six historical CLIP-initialized common-SH3 runs share the original query IDs and one fixed map; they are separate from the historical native-map comparison in Table S1. Denominators are 139 in-map and 139 out-of-map queries for Block Small, and 602 of each for Block A. Query-time quantiles cover all queries, including rejected ones, from encoding through retrieval, pose search, verification when reached, and result transfer; model/cache startup and queue waiting are excluded. Quantiles use linear interpolation between ordered query times. Within each scene, bold marks higher pose success and lower false acceptance or time, including ties. Median and 95th-percentile times are compared separately.

Scene	Reference bank	Views	Pose success ↑ 1 m/2° (%)	Pose success ↑ 5 m/5° (%)	Out-of-map accept (%) ↓	Query time (s) ↓ Median / 95th percentile
Block Small	Baseline	1,886	13.7	14.4	**0.0**	0.117 / 0.588
	Equal-size replacement	1,886	**15.8**	**16.5**	**0.0**	**0.109 / 0.587**
	Full expansion	1,945	**15.8**	**16.5**	**0.0**	**0.109** / 0.591
Block A	Baseline	13,094	**1.5**	**1.5**	**0.0**	0.125 / 0.221
	Equal-size replacement	13,094	**1.5**	**1.5**	**0.0**	**0.124 / 0.209**
	Full expansion	14,347	**1.5**	**1.5**	**0.0**	0.126 / 0.214

Table S3. **Historical RGB retrieval-stage latency.** Times are milliseconds; quantiles use linear interpolation and include rejected queries. (a) Historical SALAD-initialized geometric-v3, including retained CLIP encoding. (b) Separate CLIP-based configurations and cohorts. Timings are descriptive and are not ranked across scenes or cohorts.

(a) Nine-scene historical geometric-v3

Scene	Queries	Median (ms) ↓	95th percentile (ms) ↓
Block Small	278	53.184	72.137
Block A	1,204	101.922	122.144
NYC	360	50.939	73.183
SF	348	59.176	86.112
Citysample	284	54.040	59.646
Colosseum	252	44.233	59.522
Elvenruin	236	46.095	62.974
Park	456	59.687	86.181
Road	260	59.922	79.409
Pooled	3,678	61.076	120.157

(b) Other historical configurations

Benchmark	Initializer	Queries	Median (ms) ↓	95th percentile (ms) ↓
Nine-scene RGB, cross-camera	CLIP (historical)	3,678	34.046	75.635
Nine-scene RGB, camera-filtered	CLIP (historical)	3,678	30.529	89.236
Block A supported atlas	CLIP (historical)	200	43.817	47.198

64 candidates, first occurrences of place identifiers precede repeats. The first 20 form a shortlist returning up to five distinct places with unchanged poses, without visual verification. Offline captioning and the optional verification ablation use DeepSeek Flash at temperature zero and maximum image side 1,280 pixels. Verification checks shortlisted images without captions and ranks eligible views by relevance.

QGVI-I local features. The LoGS geometric initializer uses at most 4,096 SuperPoint keypoints, a 1,024-pixel maximum image side, non-maximum suppression radius three, and outdoor SuperGlue weights with five Sinkhorn iterations. QGVI-I returns the geometric initializer and tests distinct-keypoint support there; it does not load a Gaussian map at query time. The optional QGVI-I + photo. ablation runs the native LoGS photometric optimizer for up to 20 steps and tests support at the refined pose.

Acceptance calibration. The calibration set contains 245 in-map and 245 out-of-map queries from Block A and SF. QGVI-I uses a 343-combination search over distinct counts, inlier fraction, and image coverage. It selects successful PnP and at least 40 distinct inlier query keypoints at the initializer, with fraction and coverage thresholds zero. The rule is fixed across all nine test scenes. Matched count controls use $\{0, 4, 8, 12, 20, 40, 80\}$ and retain every validation

pose correct at either 1 m/2° or 5 m/5°. Selection maximizes out-of-map rejection, then in-map acceptance; ties favor fewer active gates and smaller thresholds. The optional QGVI-I + photo. ablation requires native final-pose eligibility and at least 40 distinct inlier query keypoints with positive depth and reprojection error at most 12 pixels at the refined pose.

C. Additional Experimental Protocol and Diagnostics

Source split and text cohorts. The localization source split contains 18,570 training and 1,309 evaluation images with no exact file overlap. Some evaluation observations contributed to reconstruction; the evaluation therefore does not test generalization to unseen reconstruction observations. In the 180-query text cohort, source-image review labels 160 descriptions fully supported, 14 partially supported, and six unsupported. All remain in the denominator. A description unsupported by its source image may still match another view. Development exposure includes 96 previously used source images and one repeated normalized description. The reported cohorts are development evaluations; no independent final evaluation has been conducted. The 180- and 135-query text cohorts use RGB test-source images; the 176-query cohort uses RGB validation-source images. The 135 descriptions are synthetic; the other two cohorts are DeepSeek-authored. Within each comparison, methods share query IDs, source splits, and target scenes. The overall text-ablation and breakdown rows pool the 176 query IDs.

RGB cohorts and cached evaluation. The 1,839 test positives comprise 480 aerial and 1,359 street views. Block A contributes 501 source images and 101 renders, versus 13 and 113 in Colosseum. The separate 988-query validation pool contains 280 source-image positives, 214 rendered positives, and 494 negatives. Out-of-map queries have no pose error and enter false-acceptance evaluation, not pose-success denominators. An always-accepting method has 100% false acceptance regardless of in-map accuracy. Initialization failures remain in the reported pose-success denominators. QGVI-I evaluation reuses frozen initializer outputs and correspondences; fresh runs on four queries reproduce those cached outputs. The optional QGVI-I + photo. ablation uses frozen native LoGS refined poses and recomputes distinct support from saved correspondences. The GSplatLoc appearance diagnostic is excluded from scored comparisons.

The 13 QGVI-I false accepts occur in Block A (2), NYC (2), SF (7), and Citysample (2). QGVI-I + photo. has 10: two in Block A, one in NYC, five in SF, and two in Citysample. Both reject the same 21 in-map queries in Block Small (5), Block A (15), and Colosseum (1). Map-training coverage also differs: the released LoGS schedule loads 3,100 of Park's 6,194 allowed cameras over 30,000 iterations; the number of distinct cameras used in optimization is unrecorded. All nine scenes are reported.

Scene-scale verification. The source-verified scales are 25 metres per model unit for Block Small and Block A, and one for NYC and SF. The latter are checked against 351 paired camera shifts with labelled 0.3 m height differences and 774 source-to-checkpoint and checkpoint-to-benchmark camera matches per scene. The other five scene scales use a calibrated ratio of DA3 raw depth to Gaussian-rendered expected camera-Z depth on frozen training street views. A conversion constant is fitted using 13 Block Small views and its verified scale, then applied unchanged to the median depth ratio in each target scene. Every method uses the same frozen scene-wide factor; it is not fitted to localization outcomes. Block A, held out from scale calibration, has a scale error of 3.9%. Calibration cameras retain their native coordinates, so scale transfer does not require map rescaling. This procedure provides a shared metric conversion for comparison, while the five DA3 estimates remain unverified by independent physical measurements.

Historical text retrieval diagnostics. All 180 queries have joint judgments; scene-macro and pooled rates agree after seed averaging. A one-query CityLoC repeatability check produced different refined poses and rankings despite fixed inputs and seed; no independent paired refinement effect has been measured. On the 135-query cohort, fusion with verification produces three Top-5-only gains and seven losses relative to caption retrieval, a net loss of four. Six losses return no result; the seventh ranks its relevant view sixth. All nine successful caption-retrieved views from these seven queries were in the 20-view shortlist. These observations do not support a tuning claim. For the 176-query shortlists, caption and CLIP success-count bounds are 152–176 and 141–175; these are not full-bank recall estimates. Returned-view validity is judged for the 135- and 180-query cohorts but unverified for the 176-query cohort. On the 180-query cohort, CityLoC refinement raises seed-averaged Top-5 success from 30.7% to 34.6%, with 19.8% no result.

Historical query timing boundaries. The earlier 176-query efficiency run uses PyTorch 2.9.1+cu128. CityLoC timings include serialization and GPU synchronization, exclude model/map startup, and use all queries with nearest-rank 95th percentiles. Peak PyTorch allocated memory includes resident models/maps but excludes driver and custom non-PyTorch allocations. The separate 135-query pilot uses 135 IDs across three seeds, including only one ID each

in NYC and SF; per-scene median times are 2.5–7.3 s without refinement and 35.6–287.6 s with it. Returning no result does not establish correct rejection.

That historical fusion + verification timing pass uses 3,562 HTTP attempts across all 176 queries and retries (20.24 per query); two query IDs end in API-response failure. The execution history contains four failed executions, including earlier retries; this count differs from the two failed query IDs. Failed executions enter neither latency quantiles nor retrieval no-result counts. Cached-response diagnostics are excluded from online timing. Provider GPU usage and memory are unknown.

D. Additional Evaluation Protocols

The following evaluations separate recovery of the description's source camera, semantic satisfaction, and query latency. The 180-query development cohort and method outputs remain fixed. QGVI-T here denotes the historical fusion variant, not the current BM25 interface.

D.1. Text Source-Camera Recovery

Each description is linked to the camera that produced its authoring image. For a predicted camera-to-world pose $(\hat{R}, \hat{c})$ and source pose $(R^\star, c^\star)$, translation error is $s\|\hat{c} - c^\star\|_2$, where s is the frozen scene scale; rotation error is the angle of $\hat{R}^\top R^\star$. Stored retrieval poses can be evaluated directly. These errors measure source-camera recovery: another view may satisfy the description without being close to its source camera.

Top-1 pose success requires both translation and rotation errors to meet the specified threshold. Pose Recall@5 requires at least one of the first five outputs to meet both thresholds; the same output must satisfy both. Empty returns fail, and shorter lists receive no additional candidates. Missing query records or invalid pose matrices invalidate an evaluation rather than becoming successful abstentions. All methods use the same query IDs, fixed ranking, and output budget. CityLoC's three seeds are averaged within each query before scene averaging.

We report the nine-scene mean and a separate mean over the four scenes with source-verified metre scales. The remaining five use the fixed DA3 estimates disclosed in Table S11. Top-1 error distributions are conditional on a returned valid pose and must be accompanied by their denominator and return rate. An auxiliary full-bank oracle tests whether any stored reference pose can satisfy the thresholds; it measures the discrete bank's coverage, not the retrieval method's accuracy. The bank covers 0/180 source cameras at 1 m/2° and 40/180 at 5 m/5°. For 171 descriptions, no bank orientation is within 2° of the source, irrespective of position. Independent checks trace all source poses to the original held-out or novel-view records and reproduce the pose scores with a second rotation implementation. This coverage bounds QGVI-T's discrete retrieval, not CityLoC's continuous pose proposals. At the looser threshold, QGVI-T retrieves 6 of the 40 queries covered by the bank within its five outputs; the optional verification variant retrieves 8.

D.2. Blinded Cross-Model Semantic Assessment

The historical CityLoC/QGVI-T comparison in Table S9 and the earlier 180- and 135-query tables retain completed Qwen/DeepSeek judgments. Qwen3-VL-8B-Instruct (revision `0c351dd0`) assigns one of four relevance labels: fully relevant, partially relevant, irrelevant, or uncertain. Separate DeepSeek image-only judgments assign render validity. Joint success requires full relevance and a valid image; uncertain labels produce lower and upper bounds. The completed Kimi/GPT assessment is reported separately in Table S5; its labels do not alter those historical Qwen/DeepSeek labels. The common first-answer text comparison reuses compatible GPT judgments and adds reviews of the frozen BM25 and QGVI-A outputs (Supplementary Sec. E).

The completed reassessment pools fixed Top-5 outputs from CityLoC with and without refinement and QGVI-T with and without verification, across all seeds on the 180-query cohort: 6,150 output occurrences yield 5,808 distinct query–pixel pairs. Identical pairs are assessed once per reviewer and their labels reused for every occurrence. Each reviewer receives only the description and rendered image; method, rank, camera pose, stored captions, and earlier judgments are withheld.

Reviewer A uses Kimi (requested model `kimi-for-coding`, Kimi Code CLI 2.2.0, low thinking effort); Reviewer B uses GPT-6.1 (requested model `gpt-6.1-sol`, Codex CLI 0.159.2, low reasoning effort). These are requested model aliases; backend version identities are not independently verified. They are called independently with the same rubric: relevance is supported, contradicted, or uncertain, and validity is valid, invalid, or uncertain. Their joint success criterion requires supported relevance and valid imagery. Their judgments and disagreements are retained separately. Failed or invalid responses were retried without replacing valid judgments. Both reviewers completed all 5,808 pairs (11,616 judgments); uncertain judgments remain unresolved. This three-way relevance rubric differs from Qwen's historical four-way rubric: a partially relevant judgment may correspond to contradiction or uncertainty. Comparing full support across the two rubrics therefore does not imply exact category agreement.

Top-1 and Top-5 success include all queries, including empty returns; View-level precision pools all returned views, whereas the primary accepted-answer precision in Sec. E uses only the first answer per query. No-result rate

Table S4. **Text source-camera recovery.** Historical fusion-based QGVI-T; same descriptions and scene-macro percentages after seed averaging. P1 is Top-1 pose success and R5 is Pose Recall@5. Error columns average conditional scene medians, with coverage reported alongside the completed results. CityLoC first averages errors over seeds returning a valid Top-1 pose within each query. The four-scene subset uses source-verified metre scales; five other scenes retain DA3 estimates. All scenes contribute conditional errors. In row order, error distributions contain 180, 160, 180, and 135 queries (80, 72, 80, and 63 in the four-scene subset); pose coverage averages seed-level returns. Bold marks highest success and lowest error within each cohort; coverage is descriptive.

Method	P1 ↑ 1 m/2°	R5 ↑ 1 m/2°	P1 ↑ 5 m/5°	R5 ↑ 5 m/5°	Mean scene median ↓ (m/°)	Pose coverage (%)
All nine scenes, 180 queries						
CityLoC w/o refinement	**0.00**	**0.00**	0.19	0.19	86.30 / 69.99	100.00
CityLoC with refinement	**0.00**	**0.00**	0.37	0.56	89.16 / 64.98	80.19
QGVI-T	**0.00**	**0.00**	**0.56**	3.33	64.66 / **51.64**	100.00
QGVI-T + verification	**0.00**	**0.00**	**0.56**	**4.44**	**47.08** / 55.26	75.00
Four scenes with source-verified scales, 80 queries						
CityLoC w/o refinement	**0.00**	**0.00**	0.42	0.42	125.34 / 68.42	100.00
CityLoC with refinement	**0.00**	**0.00**	0.83	0.83	132.09 / 62.06	80.42
QGVI-T	**0.00**	**0.00**	**1.25**	6.25	90.57 / **51.12**	100.00
QGVI-T + verification	**0.00**	**0.00**	**1.25**	**7.50**	**60.33** / 51.75	78.75

measures empty outputs without establishing whether a description is unanswerable. Full-corpus recall is unavailable because the complete relevant-view set is unknown. These are model-generated labels, and inter-model agreement alone does not establish their correctness.

Table S5 reports separate Top-1 and Top-5 scores and first-answer accounting for these four historical text arms. For A returned answers, S confirmed successes, W confirmed wrong answers, and u unresolved answers, precision lies in $[S/A, (S+u)/A]$ and wrong-return rate in $[W/N, (W+u)/N]$. The utility interval is $[(S-W-u)/N, (S+u-W)/N]$; abstention earns zero. Both judges give QGVI-T higher Top-5 lower bounds than its verified variant; their Top-1 lower-bound ordering differs. Overlapping intervals limit ranking claims. These are repeated assessments of the same development queries, not an independent test set.

D.3. Measured Query Latency

We report query latency for seven historical variants on nine scenes. These timings do not measure QGVI-A or the current BM25 interface. Matched retrieval controls have ranking-only timings; QGVI-I + photo. and calibrated baseline gates have no separate query timing. Timers exclude model and reference loading. Cached-response replay is not a fresh API query measurement.

Text queries. CityLoC with/without refinement and QGVI-T with/without verification use the same 180 descriptions. CityLoC has 540 query–seed observations per arm. Without refinement, its median/p95 is 3.376/8.389 s; with refinement, 171.720/429.821 s, with 107 empty outputs. Both arms render diffusion proposals for CLIP selection and therefore require the inherited Gaussian map. Their timers include candidate-image serialization and hashing, but exclude model/map loading. QGVI-T fusion takes 0.020/0.063 s, including diversity selection. QGVI-T + verification takes 15.465/17.067 s; all 180 queries complete, 45 return no result, and the timer includes 3,614 HTTP attempts, network round trips, and retries. Its output lists differ from the frozen accuracy predictions on 73 queries, so timing-pass outcomes are not substituted for accuracy results. QGVI-T runs on the CPU; its optional verification variant adds a remote service. The historical text runs were recorded on different dates without a matched hardware protocol; the remote service hardware is unknown.

Table S6 reports each scene separately. Each CityLoC cell pools 20 queries over three training seeds; each QGVI-T variant has 20 queries per cell. Quantiles are calculated from the individual query times, not averaged across seed or scene summaries.

Fresh RGB timing. All three RGB methods use the same 36 query IDs, selected by a fixed hash of public IDs before timing, and three resident repetitions per query. Images, calibration, models, and reference assets retain their frozen protocols. QGVI-I requires fresh features, retrieved references, matches, poses, and gate decisions to reproduce its saved implementation outputs. LoGS checks exact initializer outputs and canonical camera, image, mask, and optimizer inputs before executing the unchanged native refinement. SplatHLoc checks public input pixels and its unchanged native configuration. For the two refined baselines, differences from historical poses are retained as repeatability diagnostics.

LoGS's initial numerical check also failed when rerunning its frozen native implementation. In an eight-query diagnostic over Block Small and NYC, 7/48 direct native

Table S5. **Separate Kimi and GPT assessments of historical text methods.** All 180 development queries from nine scenes; each CityLoC variant has three seeds, giving $N = 540$ query–seed observations. Coverage and all quality rates are percentages. Top-1 success is correct-return rate Y; precision P conditions on returned answers, while wrong-return rate E divides by all queries. Net utility $100U_1$ uses per-query reward $+1$ for a correct first answer, -1 for a wrong answer, and zero for abstention. Ranges are unresolved-label bounds, not confidence intervals. Bold marks the best conservative bound within each reviewer: lower for success, precision and utility, upper for wrong returns. Overlap limits ranking claims. Native acceptance policies are retained. Historical Qwen relevance and DeepSeek validity results appear in Table S9(a).

(a) Coverage, retrieval success, and first-answer utility

Method	N	Coverage C	Top-1 success $Y \uparrow$		Top-5 success $\uparrow$		Net utility $100U_1 \uparrow$	
			Kimi	GPT	Kimi	GPT	Kimi	GPT
CityLoC w/o refinement	540	100.0	22.2–61.1	26.7–66.5	37.8–84.6	40.6–82.0	[−55.6, 22.2]	[−46.7, 33.0]
CityLoC with refinement	540	80.2	25.6–47.0	29.1–49.4	37.4–64.1	39.6–63.0	[−29.1, 13.9]	[−22.0, 18.7]
QGVI-T	180	100.0	52.2–73.9	**61.1**–77.8	**72.8**–93.3	**73.9**–91.7	[4.4, 47.8]	[22.2, 55.6]
QGVI-T + verification	180	75.0	**56.7**–68.9	60.6–68.9	67.2–74.4	68.3–72.8	[**38.3**, 62.8]	[**46.1**, 62.8]

(b) First-answer precision and wrong-return rate

Method	Answer precision $P \uparrow$		Wrong returns $E \downarrow$	
	Kimi	GPT	Kimi	GPT
CityLoC w/o refinement	22.2–61.1	26.7–66.5	38.9–77.8	33.5–73.3
CityLoC with refinement	31.9–58.7	36.3–61.7	33.1–54.6	30.7–51.1
QGVI-T	52.2–73.9	61.1–77.8	26.1–47.8	22.2–38.9
QGVI-T + verification	**75.6**–91.9	**80.7**–91.9	6.1–**18.3**	6.1–**14.4**

Table S6. **Text query latency by scene (seconds; ↓).** Median/nearest-rank p95 from the recorded query runs. CityLoC has 60 observations per cell and each QGVI-T variant has 20. Empty outputs remain included. QGVI-T uses historical fusion; verification includes remote calls and retries.

Scene	CityLoC w/o refinement	CityLoC with refinement	QGVI-T	QGVI-T + verification
Block Small	4.776 / 5.915	145.724 / 609.644	0.021 / 0.029	15.457 / 16.844
Block A	8.244 / 10.336	182.853 / 700.906	0.062 / 0.086	15.648 / 17.970
NYC	3.297 / 3.693	198.926 / 327.830	0.018 / 0.022	15.462 / 16.845
SF	2.717 / 3.660	177.899 / 347.767	0.020 / 0.024	15.334 / 16.155
Citysample	3.113 / 3.772	85.713 / 239.108	0.021 / 0.024	15.380 / 16.164
Colosseum	2.559 / 3.081	170.950 / 226.452	0.012 / 0.015	15.836 / 16.915
Elvenruin	3.411 / 3.974	115.690 / 429.821	0.011 / 0.014	15.293 / 16.075
Park	3.527 / 4.066	180.399 / 332.218	0.029 / 0.037	16.092 / 18.325
Road	3.453 / 4.046	151.034 / 257.865	0.019 / 0.027	15.077 / 17.638

calls exceeded the previous saved-pose tolerance. Identical visible forward outputs and losses produced different pose gradients. Subsequent timing therefore qualifies native source and exact external-input conformance; numerical variation is reported separately, without increasing the tolerance or selecting closer runs. This does not establish that the variation is harmless. The failed checks and fixed accuracy outputs are preserved.

For LoGS, SplatHLoc, and QGVI-I, timing starts before image decoding and ends after the pose and acceptance decision with device synchronization. Model/reference loading and qualification checks are outside the query timer. The timing subset does not estimate full-test accuracy or represent an unseen test. All local RGB geometry runs use an NVIDIA RTX PRO 6000 Blackwell Workstation Edition. LoGS retains input references and clones eight initial optimizer scalars inside the timer; input hashing and canonical-input checks follow it. PyTorch uses two CPU threads; native OpenCV threading is retained (32 threads recorded for SplatHLoc), so this is not a two-core host limit. Quantiles are computed from pooled raw observations, with nearest-rank 95th percentiles. With twelve observations per scene, this percentile is the observed maximum; this small timing sample does not establish a stable population tail.

Table S7. **RGB latency by scene (seconds; ↓).** Median/nearest-rank p95 of twelve resident observations per method and scene. Bold marks the lowest displayed median and p95 across three qualified methods, including ties. QGVI-I is the geometric default; QGVI-I + photo. has no matched timing.

Scene	LoGS	SplatHLoc	QGVI-I
Block Small	0.487 / 0.789	0.529 / 0.630	**0.286 / 0.342**
Block A	0.425 / 0.722	0.527 / **0.544**	**0.329** / 0.647
NYC	1.065 / 2.732	0.399 / 0.877	**0.193 / 0.246**
SF	0.365 / 0.776	**0.107** / 0.857	0.232 / **0.300**
Citysample	0.343 / 1.424	0.543 / 0.649	**0.198 / 0.262**
Colosseum	0.318 / 0.366	**0.118** / 0.613	0.227 / **0.287**
Elvenruin	0.564 / 2.305	0.530 / 0.658	**0.222 / 0.334**
Park	0.508 / 0.979	0.385 / 0.735	**0.229 / 0.307**
Road	0.459 / 0.554	**0.108** / 0.588	0.259 / **0.453**

E. Shared Answer Accounting and Net Query Utility

The query routes return answers whose correctness can be evaluated, but their correctness predicates differ. For text, the first returned view must be usable and satisfy every required semantic constraint. For RGB, the query must be in-map and the returned pose must satisfy both geometric thresholds. An accepted out-of-map image or an inaccurate in-map pose is incorrect. For image-and-instruction queries, the output view must be usable and satisfy every required target predicate, including its relation to the source image; recovering the source camera alone is insufficient. This shared accounting follows selective prediction [15]; it does not equate semantic satisfaction with geometric accuracy.

For N queries, let $a_i = 1$ when query i receives a first answer and $z_i = 1$ when that answer is correct. Set $a_i z_i = 0$ for an empty return. With $A = \sum_i a_i$, $S = \sum_i a_i z_i$, and $W = A - S$, define

$$C = \frac{A}{N}, \qquad P = \frac{S}{A}, \qquad Y = \frac{S}{N} = CP, \qquad E = \frac{W}{N}. \tag{5}$$

Coverage C measures how often the method answers; accepted-answer precision P measures how often those answers are correct; correct-return rate Y measures how often a query receives a correct answer. Wrong-return rate E counts incorrect answers over all queries. Selective risk is $1 - P$. If $A = 0$, precision and selective risk are undefined, while $C = Y = E = 0$. The identity uses fractions and pooled counts; it need not hold after independently averaging scene-level C and P.

Explicit costs for wrong answers. Cost-based evaluation makes the relative consequences of answering and abstaining explicit [14]. We assign reward $+1$ to a correct first answer, reward $-\lambda$ to an incorrect first answer, and zero to abstention:

$$U_\lambda = \frac{S - \lambda W}{N} = Y - \lambda E, \qquad \lambda \geq 0. \tag{6}$$

An all-reject system scores zero; a system returning only wrong answers scores negatively for $\lambda > 0$. The identities $U_0 = Y$ and $U_1 = 2Y - C$ show that utility adds an explicit preference over the existing counts, rather than additional evidence. We report $100U_\lambda$ in net reward units per 100 queries, not as an accuracy percentage. Values can fall below -100 when $\lambda > 1$.

The cost grid $\lambda \in \{0, 0.5, 1, 2, 5\}$ is a descriptive sensitivity analysis specified after the recorded evaluations. It was not used to calibrate gate thresholds. The default designation follows review of these exposed development results. Equal reward and penalty ($\lambda = 1$) provide an interpretable example; no deployment cost has been established. This standard utility construction is an evaluation choice, not a methodological contribution.

Table 1 uses frozen localization outputs. Panel (a) applies one GPT-low relevance/validity rubric to text; panel (c) retains separate high-effort predicate judgments for image-and-instruction outputs. CityLoC contributes three equally weighted seed runs per query. RGB uses a balanced in-map/out-of-map population and the 1 m/2° threshold, including the declared estimated-scale scenes. The maximum possible RGB correct-return rate is 50% and its maximum U_λ is 0.5 in this mixture because out-of-map queries have no correct pose. Rejection receives no positive reward. Consequently, text and RGB panels must not be ranked against each other or pooled into one score. The in-map pose-success rate and out-of-map rejection rate remain separate diagnostics in Table S11.

Common low-effort text assessment. All 180 descriptions are evaluated with the same GPT-6.1 Sol low-effort relevance/validity prompt and label schema. Relevance and render usability are judged from the unchanged query and output image, without method identity or controller rationale. Historical labels are reused only when query, image bytes, requested model, effort, prompt, and schema match. New reviews use native Codex sessions; historical reviews used the Codex CLI with ChatGPT subscription authentication. These transports and service dates differ, so matching the requested configuration does not guarantee identical backend execution. The first unverified BM25 result is recovered from the frozen QGVI-A prefix; original-image cameras are evaluated through their cached Gaussian renders. Every method retains all 180 descriptions (three seeds for CityLoC), including nonreturns. Table S8 reports all six methods; historical fusion and verification are not omitted from the comparison. No missing review is treated as an incorrect answer. An uncertain label in either dimension leaves the joint result unresolved, even when the other

dimension is negative. These judgments remain intervals; they are not resolved by selecting a judge.

Agentic evaluation protocol. QGVI-A uses GPT-6.1 Sol, low effort. Prefix and continuation share limits of 12 decisions (including actions, transitions, and finish) and six render attempts; failures consume these budgets. QGVI-T's BM25 interface searches original-image and grid captions, returning up to six candidates without embeddings. The RGB tool uses QGVI-I's geometric initializer and distinct-support gate. These runs do not evaluate the optional photometric refinement inside QGVI-A. Calls are query-dependent; an episode need not invoke both tools. Camera-local moves and bounded pose proposals produce Gaussian renders without traversability constraints. The prefix checks geometric support for source poses, or rendered target predicates and source identity when supplied. The 250 image-and-instruction queries use 125 source images: 32 queries in Block A, 22 in Elvenruin, and 28 in each other scene. Text and RGB retain 180 and 3,678 queries. RGB reuses the historical pixels, calibration, membership, scales, and thresholds; newly executed poses need not be identical.

The earlier semantic assessment used a fresh high-effort request from the same model family, receiving the instruction, required predicates, and final output render; image-and-instruction reviews also receive the source image. Review images have a maximum side of 512 pixels. The reviewer does not receive the controller's rationale. An unusable render or a violated predicate is a failure; otherwise, any unresolved predicate yields an uncertain judgment. These are model assessments, not human ground truth. This high-effort protocol remains the target-view evaluation and a text sensitivity result. The new common low-effort assessment supplies the matched text comparison; the two protocols are not pooled.

Observed trade-offs. Under the historical Qwen/DeepSeek rubric, at $\lambda = 1$ fusion yields 16.67 net reward units per 100 queries, versus $-50.00/-27.96$ for CityLoC. Verification yields 57.22 and dominates raw fusion for every $\lambda \geq 0$: 119 versus 105 correct first answers, and 16 versus 75 wrong answers. It sacrifices Top-5 success and latency (Tables S20, 5). Raw fusion's utility becomes negative for $\lambda > 1.4$.

QGVI-I yields 46.74 per 100 RGB queries with 56 wrong returns: 43 inaccurate in-map poses and 13 out-of-map accepts. QGVI-I + photo. yields 46.82 with 53 wrong returns (43 and ten); their utility curves nearly coincide. Native LoGS/SplatHLoc yield $-3.48/-38.36$. Figure 3 shows pooled and equal-scene utility versus error cost. Both distinct-support thresholds are 40; SplatHLoc's retention constraint selects its native count gate (Table 2; Supplementary Sec. K).

Under the earlier high-effort protocol, QGVI-A returns 131 answers to 180 text queries: 125 are judged correct, three incorrect, and three uncertain; 49 queries have no return. On the 250 target-view queries, it returns 244 answers: 226 correct, 17 incorrect, and one uncertain. The six nonreturns include one invalid model-output outcome. Table 1(c) now compares these fixed outputs with two completed controls on the same target-view cohort; their policies, budgets, and paired comparisons are detailed in Sec. F. On RGB, QGVI-A returns 1,831 poses, including 1,773 strict successes, 46 inaccurate in-map poses, and 12 out-of-map accepts. Its net utility is 46.63, below QGVI-I's 46.74 and QGVI-I + photo.'s 46.82. The 20 rejected in-map queries give a false-rejection rate of 1.088%; the 12 accepted out-of-map queries give a false-acceptance rate of 0.653%.

Output budgets and unresolved labels. The shared primary definition uses one answer. Text Top-5 success remains a separate fixed-budget measure: at least one of at most five returned views must be correct. It does not imply that all five views are correct. If L accepted answers are labelled correct and U have unresolved labels, precision lies in $[L/A, (L+U)/A]$ and correct-return rate in $[L/N, (L+U)/N]$. Utility lies in

$$\left[\frac{(1+\lambda)L - \lambda A}{N}, \frac{(1+\lambda)(L+U) - \lambda A}{N}\right]. \quad (7)$$

These are label-uncertainty bounds, not sampling intervals. The matched low-effort text bounds retain every unknown judgment; no reviewer responses are missing. The earlier high-effort text and target-view protocols retain three and one uncertain judgments, respectively. CityLoC's 540 observations are three runs on 180 query IDs; they are not 540 independent queries. Missing reviewer responses do not become negative labels. Runtime failures count as unsuccessful queries and are reported separately from deliberate abstention; missing execution records invalidate an incomplete evaluation.

Acceptance and answerability. For a confidence score, thresholds should be selected on validation data under a common retention or risk objective, then frozen for test evaluation. Risk–coverage curves require a recorded score and an explicit threshold policy; isolated binary decisions provide operating points, not a continuous curve. The text cohort does not establish whether each description is answerable anywhere in the scene. We therefore do not credit an empty text return as correct rejection or report text false-accept/false-reject rates. An unsupported source image does not imply that all other views fail. Likewise, Y is not answerability recall, so combining P and Y into a harmonic mean would not produce a conventional F1 score. Full shared risk–coverage curves cannot be recovered from the

Table S8. **Complete common-rubric text comparison.** All six frozen methods use GPT-6.1 Sol, low effort, and the same relevance/validity rubric. CityLoC has three seeds per description; other methods have one output list. Ranges retain uncertain labels. Bold marks the best conservative endpoint (lower for precision, correct returns and utility; upper for wrong returns), including ties; this does not establish statistical superiority.

Method	N	C (%)	P (%) ↑	Y (%) ↑	E (%) ↓	$100U_1$ ↑
CityLoC w/o refinement	540	100.00	26.67–66.48	26.67–66.48	33.52–73.33	-46.67–32.96
CityLoC with refinement	540	80.19	36.26–61.66	29.07–49.44	30.74–51.11	-22.04–18.70
QGVI-T	180	100.00	60.00–70.56	60.00–70.56	29.44–40.00	20.00–41.11
QGVI-A	180	72.78	**90.84–93.89**	**66.11–68.33**	**4.44–6.67**	**59.44–63.89**
Historical fusion	180	100.00	61.11–77.78	61.11–77.78	22.22–38.89	22.22–55.56
Fusion + verification	180	75.00	80.74–91.85	60.56–68.89	6.11–14.44	46.11–62.78

Table S9. **Historical text assessment protocols.** These saved operating points retain their original Qwen/DeepSeek or high-effort GPT judgments; they are not the matched low-effort comparison in Table 1. Bold is retained from the original within-panel comparison.

Method	N	C (%)	P (%) ↑	Y (%) ↑	E (%) ↓	$100U_1$ ↑
(a) Historical Qwen/DeepSeek rubric						
CityLoC w/o refinement	540	100.00	25.00	25.00	75.00	-50.00
CityLoC with refinement	540	80.19	32.56	26.11	54.07	-27.96
QGVI-T (fusion)	180	100.00	58.33	58.33	41.67	16.67
Fusion + verification	180	75.00	**88.15**	**66.11**	**8.89**	**57.22**
(b) Earlier QGVI-A high-effort predicate rubric						
QGVI-A	180	72.78	95.42–97.71	69.44–71.11	1.67–3.33	66.11–69.44

available operating points: some native abstentions have no returned answer whose correctness can be scored when hypothetically relaxing the policy.

F. Target-View Retrieval and Fixed-Search Controls

We compare frozen QGVI-A outputs with two controls on the same 250 image-and-instruction queries and 125 source images. Required predicates, source pixels, intrinsics, scene assets, and the final assessment rubric are held fixed. Each source contributes two tasks. This is an exposed development cohort, not an independent final test.

Control policies. *Retrieve + Select* renders the initial six BM25 candidates using the task camera calibration. In one GPT-6.1 Sol low-effort call, the model sees the source and all available candidate renders and either selects one supported existing view or reports insufficient evidence. It cannot move a camera. *Fixed Search + Verify* starts from the first BM25 candidate and inspects six precommitted poses: the anchor, yaw left/right by 10°, then translate left/right/backward by a training-camera-derived step. All offsets are relative to the anchor. The same low-effort model checks one view at a time and stops at the first fully supported answer; it cannot choose movements, inspect future views, or see prior judgments. This deterministic control does not reproduce a separately published method. Candidate cameras were frozen before new model outcomes; frozen adaptive answers were not used to choose their movements. Exact renders are reused only when camera, calibration, dimensions, renderer binding, and pixel hashes agree; other candidates use the unchanged Gaussian renderer.

Budgets and assessment. Each control allows six render slots; invalid or out-of-bounds slots are not replaced. Retrieve + Select uses 250 model calls and 1,500 slots in total (means 1.000 and 6.000). Fixed Search + Verify uses 1,291 calls and 1,326 inspected slots (means 5.164 and 5.304). QGVI-A instead has shared prefix/continuation limits of 12 decisions and six render attempts, with failures consuming the budget. Decisions include actions, transitions, and finish; they are not equivalent to model calls. The selected final adaptive executions use 991 calls (mean 3.964, range 1–10), excluding independent reviews and discarded recovery attempts. The adaptive policy can also inspect captions, reference photographs, history, and crops, and use PnP or continuous pose proposals. The controls do not receive these same observations or tools. Thus information access, feedback, stopping, and computation differ: the comparison tests complete policies and does not isolate navigation as a causal component.

Every returned image receives the same source-aware

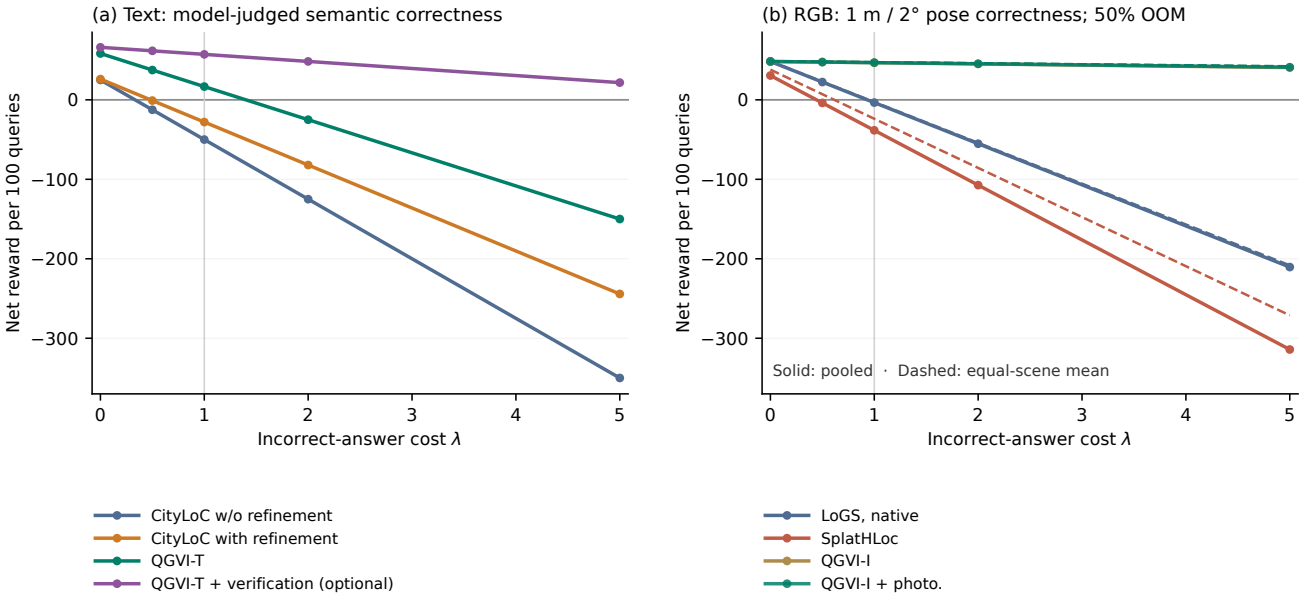


Figure 3. **Sensitivity to the cost of an incorrect first answer.** Correct answers earn $+1$, incorrect answers earn $-\lambda$, and abstentions earn zero. Curves connect the declared cost grid and are linear by definition. Solid curves use pooled query counts; dashed RGB curves weight scenes equally. Text aggregation schemes coincide. Text curves show historical fusion with Qwen/DeepSeek labels. The horizontal line is all-reject utility. The two modalities have different correctness definitions and query populations; their heights must not be compared. LoGS and SplatHLoc retain native acceptance; QGVI-I gates initializer poses, and QGVI-I + photo. gates refined final poses. Curves use frozen outputs, without new inference or threshold optimization.

GPT-6.1 Sol high-effort assessment used for the frozen adaptive output. The reviewer sees only the source, final render, instruction, required predicates, and schema; method, controller reasoning, and other judgments are withheld. This is an independent call within the same model family, not human ground truth. Exact matched historical assessments are reused and identical new review inputs are deduplicated. Different session transports and service dates, together with separately prepared shared-GPU renders, preclude a controlled latency comparison; no runtime ranking follows from these records.

Counts and paired uncertainty. All 750 executions and all 362 returned-view assessments are complete. Retrieve + Select returns 68 answers (67 correct, one wrong); Fixed Search + Verify returns 50 (44 correct, five wrong, one uncertain); QGVI-A returns 244 (226 correct, 17 wrong, one uncertain). Their no-return counts are 182, 200, and six, respectively. The adaptive six include one invalid model-output failure; it is retained as a zero-utility nonreturn, separately from deliberate abstention. No missing execution or assessment is scored as a semantic failure.

Relative to QGVI-A, the controls' correct-return differences have label-uncertainty bounds of $[-64.00, -63.60]$ and $[-73.20, -72.40]$ percentage points. Their utility differences are $[-57.60, -56.80]$ and $[-68.80, -67.20]$ net reward units per 100 queries. These bounds enumerate unresolved labels and are not confidence intervals. Separately, a paired percentile bootstrap resamples the 125 source-image clusters, retaining both queries per source (5,000 resamples; fixed seed 20261102). The confirmed-success differences are -63.60 and -72.80 points, with 95% bootstrap intervals $[-70.80, -55.60]$ and $[-79.20, -66.00]$. This statistic counts confirmed labels without imputing uncertain answers; its sampling interval does not resolve label uncertainty or establish generalization beyond the development cohort. Table S10 shows the same accounting by scene and task family. Higher answer yield does not imply uniformly higher precision: Retrieve + Select confirms 67 of 68 returned answers, while abstaining on 182 queries.

G. Nine-Scene Retrieval and Localization Details

Table S11 retains the per-scene text Top-1/Top-5 success, no-result rates, RGB pose success, false-accept/false-reject rates, and scale factors. These diagnostics complement the pooled first-answer accounting in main Table 1; text rows retain historical fusion and Qwen/DeepSeek judgments, rather than the common GPT-low assessment. The two panels use their original query populations and metric denominators; scene means weight all nine scenes equally.

Table S10. **Target-view controls by scene and task family.** All methods use the same queries within each row. $S/W/u$ are counts of confirmed correct, wrong, and uncertain returns; the remaining $N - S - W - u$ queries have no return. Utility $100U_1$ uses every query. Ranges reflect semantic-label uncertainty, not sampling intervals. Bold marks a utility interval lying strictly above both alternatives. Policies and actual budgets differ; these results do not isolate a navigation effect.

		Retrieve + Select		Fixed Search + Verify		QGVI-A	
Group	N	$S/W/u$	$100U_1 \uparrow$	$S/W/u$	$100U_1 \uparrow$	$S/W/u$	$100U_1 \uparrow$
Block Small	28	6/0/0	21.43	2/0/0	7.14	27/1/0	**92.86**
Block A	32	5/0/0	15.62	3/1/0	6.25	31/1/0	**93.75**
NYC	28	7/0/0	25.00	3/1/0	7.14	23/2/1	**71.43–78.57**
SF	28	11/0/0	39.29	14/0/0	50.00	25/3/0	**78.57**
Citysample	28	7/0/0	25.00	5/0/0	17.86	25/1/0	**85.71**
Colosseum	28	14/0/0	50.00	10/2/0	28.57	25/2/0	**82.14**
Elvenruin	22	2/0/0	9.09	3/1/1	4.55–13.64	18/3/0	**68.18**
Park	28	11/1/0	35.71	2/0/0	7.14	25/3/0	**78.57**
Road	28	4/0/0	14.29	2/0/0	7.14	27/1/0	**92.86**
Multi-object composition	80	12/0/0	15.00	6/1/0	6.25	69/9/0	**75.00**
Relation disambiguation	70	25/0/0	35.71	14/3/0	15.71	63/4/1	**82.86–85.71**
Visibility / framing	100	30/1/0	29.00	24/1/1	22.00–24.00	94/4/0	**90.00**

H. Optional Text Verification

Historical QGVI-T + verification evaluates the 20 fusion-shortlisted views using a VLM. The verifier receives the rendered image and query, excluding the caption. It enumerates mandatory constraints, including conjunctions, counts, attributes, and spatial relations, and assigns each a label with supporting visual evidence: supported, contradicted, or unclear. A view qualifies only if the verifier reports a match and every reported requirement is supported. Response validation overrides a match verdict when any requirement is contradicted or unclear. Contradiction produces a non-match decision; insufficient evidence produces an uncertain decision.

Eligible views are ranked by verifier relevance, with shortlist order breaking ties. Scores are not calibrated probabilities. We return at most five distinct places with unchanged stored poses, or no result when none qualifies. This abstention concerns the shortlist and does not establish absence throughout the scene. Query processing uses cached renders without new rendering calls.

I. Matched Text Retrieval Controls

CLIP, BM25, MiniLM, and historical QGVI-T fusion use the same 30,522 rendered views, 180 descriptions, and place selection: 64 candidates, a 20-view shortlist, and five distinct returned places. The three caption methods share the same captions; all four omit verification. Each reviewer judges identical query–image pairs once, using the rubric in Sec. D.2.

J. View-Bank Composition and Novel Views

We vary the candidate observations while retaining the same 180 development descriptions, historical fusion scorer, and five-place output budget. The original-image bank contains 18,570 training images, the grid bank contains 30,522 Gaussian renders, and their union contains 49,092 images. All returned cameras are assessed using their Gaussian renderings, including cameras retrieved through original-image captions. Thus the comparison measures semantic access to the supplied asset.

GPT completed all 3,263 distinct query–image judgments for the 64 bank conditions and all 1,329 judgments for the 18 original/union conditions. The following tables use this completed panel. Kimi returned 3,023/3,263 and 1,200/1,329 valid judgments, respectively; the remaining 240 and 129 pairs exhausted the retry budget. These incomplete panels are not used to assert a second-reviewer confirmation of the bank comparisons. Uncertain labels remain unresolved even in the completed GPT panel.

On all nine scenes, adding the grid changes confirmed Top-5 success from 132/180 to 134/180. Its paired change lies in $[-10.0, +12.2]$ percentage points under unresolved labels. This result does not establish a consistent improvement from novel-view augmentation.

J.1. Appearance and Candidate Coverage

The controlled study uses Block A and SF. A same-pose bank replaces each training image by a Gaussian rendering at the identical camera. The expanded-render bank adds the grid to these training-camera renders. It differs from the nine-scene original-plus-grid bank, which retains the original training images. Table S14 reports both the appearance

Table S11. **Text retrieval and image localization across all nine scenes.** Percentages; scene means weight all nine scenes equally. (a) Joint relevance/validity success under historical Qwen/DeepSeek judgments on 180 development descriptions (20 per scene); CityLoC averages three seeds; QGVI-T uses historical fusion. Empty returns fail. (b) Pose success and false rejection use all in-map queries; false acceptance uses all out-of-map queries. LoGS and SplatHLoc use native maps. QGVI-I gates the LoGS geometric initializer without photometric refinement; the gate is calibrated on Block A/SF validation, then fixed. Baselines retain their acceptance rules. † DA3-estimated scale; others source-verified. Scales are fixed across methods and all scenes enter the means. Development results; bold marks the best comparable rate and displayed ties. No-result rates are descriptive.

(a) Text-conditioned view retrieval

Method	Metric (%)	Block Small	Block A	NYC	SF	Citysample	Colosseum	Elvenruin	Park	Road	Scene mean
Queries	n	20	20	20	20	20	20	20	20	20	—
CityLoC [26] without refinement	Top-1 success ↑	21.7	15.0	21.7	43.3	30.0	33.3	20.0	20.0	20.0	25.0
	Top-5 success ↑	25.0	21.7	21.7	45.0	33.3	40.0	23.3	35.0	31.7	30.7
	No result	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CityLoC [26] with refinement	Top-1 success ↑	21.7	15.0	16.7	48.3	35.0	38.3	20.0	16.7	23.3	26.1
	Top-5 success ↑	25.0	18.3	30.0	56.7	43.3	50.0	25.0	31.7	31.7	34.6
	No result	26.7	35.0	15.0	1.7	41.7	8.3	40.0	8.3	1.7	19.8
QGVI-T	Top-1 success ↑	**65.0**	**60.0**	**60.0**	**55.0**	**50.0**	**50.0**	**80.0**	**40.0**	**65.0**	**58.3**
	Top-5 success ↑	**90.0**	**80.0**	**85.0**	**85.0**	**65.0**	**80.0**	**90.0**	**55.0**	**75.0**	**78.3**
	No result	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

(b) RGB image-query camera localization

Method	Metric (%)	Block Small	Block A	NYC	SF	Citysample	Colosseum	Elvenruin	Park	Road	Scene mean
In-map queries	n	139	602	180	174	142	126	118	228	130	—
Out-of-map queries	n	139	602	180	174	142	126	118	228	130	—
Scale	m / native unit	25	25	1	1	24.777†	12.287†	14.135†	1.040†	1.339†	—
LoGS [8]	Pose 1 m/2° ↑	**90.6**	**93.4**	**100.0**	**100.0**	**100.0**	**91.3**	**100.0**	**100.0**	**100.0**	**97.3**
	Pose 5 m/5° ↑	**92.1**	**93.5**	**100.0**	**100.0**	**100.0**	**91.3**	**100.0**	**100.0**	**100.0**	**97.4**
	False accept ↓	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
	False reject ↓	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
SplatHLoc [41]	Pose 1 m/2° ↑	65.5	12.6	**100.0**	**100.0**	62.7	82.5	91.5	84.6	84.6	76.0
	Pose 5 m/5° ↑	69.1	18.1	**100.0**	**100.0**	78.9	83.3	94.9	88.6	87.7	80.1
	False accept ↓	100.0	98.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	99.8
	False reject ↓	**0.0**	0.8	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	0.1
QGVI-I	Pose 1 m/2° ↑	**90.6**	**93.4**	**100.0**	**100.0**	**100.0**	**91.3**	**100.0**	**100.0**	**100.0**	**97.3**
	Pose 5 m/5° ↑	**92.1**	**93.5**	**100.0**	**100.0**	**100.0**	**91.3**	**100.0**	**100.0**	**100.0**	**97.4**
	False accept ↓	**0.000**	**0.332**	**1.111**	**4.023**	**1.408**	**0.000**	**0.000**	**0.000**	**0.000**	**0.764**
	False reject ↓	3.597	2.492	**0.000**	**0.000**	**0.000**	0.794	**0.000**	**0.000**	**0.000**	0.765

change and the subsequent coverage change.

J.2. Replacement and Density Controls

Replacement substitutes a fraction of the training-camera renders with grid renders while retaining the original image count and domain counts. Growth retains every training-camera render and adds a fraction of the eligible grid. Both use three fixed geometry-based sampling seeds. NYC and SF have no aerial grid; their original aerial observations remain in the nine-scene comparison. Within the two-scene replacement study, this constraint limits SF to actual replacement fractions of 11.9%, 23.7%, and 35.4% at nominal settings of 25%, 50%, and 75%.

All paired Top-5 intervals in these controls include zero. The variation across fractions and seeds does not support a monotonic retrieval benefit from a denser bank. These observations concern the sampled banks and the exposed development descriptions.

J.3. Paired Results by Description Source

The descriptions were authored from 74 held-out original dataset images and 106 novel Gaussian renders. Original dataset images include both real and synthetic source datasets. The source groups have different scene mixtures: Park contributes 18 original-source and two render-source queries, whereas Colosseum contributes one and 19. The source-stratified differences therefore do not isolate a causal effect of query origin.

Table S12. **Matched text retrieval by scene.** (a) GPT joint relevance-and-validity Top-1/Top-5 success bounds (%), with 20 queries per scene and 180 pooled. GPT completed all 2,825 distinct query–image assessments. (b) Separate pooled Kimi assessment: 2,660/2,825 judgments completed; 165 missing judgments remain unresolved. All methods return five places for every query. Intervals include uncertain labels and, for Kimi, missing judgments; they are not confidence intervals. Bold marks the highest conservative lower endpoint per metric and row in (a), and per metric in (b), including ties. Overlap limits ranking claims.

(a) GPT assessment

Scene	CLIP		BM25		MiniLM		QGVI-T (fusion)	
	Top-1 ↑	Top-5 ↑	Top-1 ↑	Top-5 ↑	Top-1 ↑	Top-5 ↑	Top-1 ↑	Top-5 ↑
Block Small	35.0–40.0	55.0–70.0	60.0–75.0	70.0–95.0	60.0–70.0	**80.0**–90.0	**65.0**–70.0	**80.0**–90.0
Block A	30.0–60.0	45.0–70.0	**60.0**–85.0	**70.0**–95.0	55.0–65.0	60.0–80.0	50.0–60.0	65.0–85.0
NYC	40.0–55.0	75.0–100.0	**65.0**–80.0	**80.0**–95.0	55.0–75.0	75.0–95.0	**65.0**–85.0	70.0–95.0
SF	60.0–80.0	85.0–100.0	65.0–80.0	90.0–95.0	**80.0**–90.0	90.0–95.0	70.0–90.0	**95.0**–100.0
Citysample	45.0–65.0	60.0–80.0	45.0–75.0	**75.0**–85.0	50.0–75.0	50.0–80.0	**55.0**–80.0	65.0–90.0
Colosseum	40.0–50.0	55.0–85.0	**75.0**–85.0	**85.0**–90.0	55.0–75.0	75.0–95.0	60.0–80.0	80.0–90.0
Elvenruin	50.0–70.0	85.0–90.0	**85.0**–90.0	**95.0**–100.0	75.0–85.0	**95.0**–100.0	**85.0**–95.0	**95.0**–100.0
Park	30.0–40.0	45.0–55.0	**45.0**–60.0	**50.0**–75.0	40.0–65.0	**50.0**–80.0	35.0–55.0	40.0–75.0
Road	25.0–30.0	55.0–75.0	**65.0**–80.0	**80.0**–100.0	40.0–65.0	60.0–95.0	**65.0**–85.0	75.0–100.0
Pooled	39.4–54.4	62.2–80.6	**62.8**–78.9	**77.2**–92.2	56.7–73.9	70.6–90.0	61.1–77.8	73.9–91.7

(b) Kimi assessment, pooled over the same 180 queries

Method	Top-1 (%) ↑	Top-5 (%) ↑	Net utility $100U_1$ ↑
CLIP	32.2–57.2	57.8–90.6	$[-35.6, 14.4]$
BM25	**55.6**–75.6	**75.6**–96.1	$[\mathbf{11.1}, 51.1]$
MiniLM	50.6–76.7	66.1–93.9	$[1.1, 53.3]$
QGVI-T (fusion)	52.2–73.9	72.8–93.3	$[4.4, 47.8]$

Table S13. **Candidate-bank comparison by scene (GPT).** Each scene has 20 queries; pooled rows use all 180. Entries are lower–upper success bounds in percent, not confidence intervals. Top-1 requires a successful first result; Top-5 requires at least one successful result among five. Bold marks the highest confirmed-success lower bound in each scene and metric, including ties; overlapping bounds limit ranking claims. The original bank uses training images, and the union retains those images while adding the grid.

Scene	Original images		Grid renders		Original + grid	
	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5
Block Small	50.0–50.0	75.0–85.0	**65.0**–70.0	**80.0**–90.0	55.0–55.0	75.0–80.0
Block A	40.0–65.0	**70.0**–85.0	**50.0**–60.0	65.0–85.0	45.0–70.0	**70.0**–90.0
NYC	**65.0**–95.0	**75.0**–95.0	**65.0**–85.0	70.0–95.0	**65.0**–90.0	**75.0**–95.0
SF	75.0–85.0	85.0–100.0	70.0–90.0	**95.0**–100.0	**80.0**–90.0	85.0–100.0
City Sample	**55.0**–80.0	**65.0**–90.0	**55.0**–80.0	**65.0**–90.0	**55.0**–80.0	**65.0**–95.0
Colosseum	**65.0**–80.0	**85.0**–90.0	60.0–80.0	80.0–90.0	**65.0**–80.0	80.0–95.0
Elven Ruin	70.0–75.0	80.0–95.0	**85.0**–95.0	**95.0**–100.0	80.0–85.0	**95.0**–95.0
Park	**45.0**–60.0	**55.0**–85.0	35.0–55.0	40.0–75.0	**45.0**–65.0	**55.0**–85.0
Road	55.0–70.0	70.0–90.0	**65.0**–85.0	**75.0**–100.0	50.0–75.0	70.0–90.0
Pooled	57.8–73.3	73.3–90.6	**61.1**–77.8	73.9–91.7	60.0–76.7	**74.4**–91.7

Confirmed Top-5 successes change from 53 to 52 in the original-source group and from 79 to 82 in the render-source group. Changes in confirmed counts can involve uncertain outcomes, so they need not equal W minus L. Both source groups retain paired uncertainty intervals containing zero.

K. Matched rejection controls and photometric refinement

We apply alternative acceptance rules to the same saved poses and calibrate each rule using only the 490 Block A/SF validation queries (245 in-map and 245 out-of-map). Count thresholds are $\{0, 4, 8, 12, 20, 40, 80\}$; ratio thresholds are $\{0, .05, .1, .2, .35, .5, .7\}$. A feasible rule must retain every pose in each validation-correct set at $1\,\mathrm{m}/2^\circ$ and $5\,\mathrm{m}/5^\circ$.

Table S14. **Same-pose appearance and expanded-render controls (GPT).** Block A and SF have $N = 20$ queries each; pooled rows have $N = 40$. Same-pose uses Gaussian renders at all training cameras; expanded-render adds grid renders. Bounds are label-uncertainty intervals. Changes are paired Top-5 differences in percentage points against the original-image bank (O) and same-pose bank (R). Bold compares the two rendered-bank conditions within each cohort.

Cohort	Candidate bank	Images	Top-1 (%)	Top-5 (%)	Δ Top-5 vs O	Δ Top-5 vs R
Block A	Same-pose renders	5,138	50.0–65.0	65.0–85.0	[−20.0, +15.0]	[+0.0, +0.0]
Block A	Expanded renders	18,232	**55.0**–75.0	**70.0**–90.0	[−15.0, +20.0]	[−5.0, +15.0]
SF	Same-pose renders	666	**75.0**–85.0	**90.0**–95.0	[−10.0, +10.0]	[+0.0, +0.0]
SF	Expanded renders	3,549	**75.0**–85.0	**90.0**–95.0	[−10.0, +10.0]	[−5.0, +5.0]
Pooled	Same-pose renders	5,804	62.5–75.0	77.5–90.0	[−15.0, +12.5]	[+0.0, +0.0]
Pooled	Expanded renders	21,781	**65.0**–80.0	**80.0**–92.5	[−12.5, +15.0]	[−5.0, +10.0]

Table S15. **All replacement and growth settings (GPT).** Every row averages all three fixed seeds on the same 20 queries; seeds do not increase the query denominator. Success columns average the lower and upper label-uncertainty endpoints. The seed range is the minimum–maximum confirmed Top-5 rate. Δ gives mean paired Top-5 uncertainty endpoints against the same-pose bank, in percentage points. For replacement, actual fraction is the proportion of training entries replaced; for growth, it is the proportion of eligible grid entries added. Bold marks the largest mean confirmed rate within each scene and policy.

Scene	Policy	Nominal (%)	Actual (%)	Images	Mean Top-1 bounds (%)	Mean Top-5 bounds (%)	Top-5 seed range (%)	Δ Top-5
Block A	Replace	0	0.0	5,138	50.0–65.0	65.0–85.0	65.0–65.0	[+0.0, +0.0]
Block A	Replace	25	25.0	5,138	55.0–65.0	68.3–85.0	65.0–70.0	[−13.3, +13.3]
Block A	Replace	50	50.0	5,138	**56.7**–70.0	68.3–88.3	65.0–75.0	[−15.0, +20.0]
Block A	Replace	75	75.0	5,138	48.3–61.7	**71.7**–91.7	70.0–75.0	[−13.3, +26.7]
Block A	Grow	25	25.0	8,412	50.0–66.7	68.3–86.7	65.0–70.0	[−5.0, +10.0]
Block A	Grow	50	50.0	11,685	51.7–68.3	68.3–86.7	65.0–70.0	[−6.7, +10.0]
Block A	Grow	100	100.0	18,232	**55.0**–75.0	**70.0**–90.0	70.0–70.0	[−5.0, +15.0]
SF	Replace	0	0.0	666	75.0–85.0	90.0–95.0	90.0–90.0	[+0.0, +0.0]
SF	Replace	25	11.9	666	78.3–85.0	90.0–95.0	90.0–90.0	[+0.0, +0.0]
SF	Replace	50	23.7	666	**83.3**–88.3	90.0–95.0	90.0–90.0	[+0.0, +0.0]
SF	Replace	75	35.4	666	76.7–85.0	**91.7**–98.3	90.0–95.0	[+0.0, +5.0]
SF	Grow	25	25.0	1,387	71.7–83.3	**90.0**–95.0	90.0–90.0	[−3.3, +3.3]
SF	Grow	50	50.0	2,108	71.7–83.3	**90.0**–95.0	90.0–90.0	[−5.0, +5.0]
SF	Grow	100	100.0	3,549	**75.0**–85.0	**90.0**–95.0	90.0–90.0	[−5.0, +5.0]

Table S16. **Paired original-plus-grid versus original-only outcomes (GPT).** W/L/S/U denote confirmed wins, confirmed losses, unchanged outcomes, and unresolved changes. Identical unresolved answer sets have zero paired change and enter S. Bounds are possible paired changes under unresolved labels, not confidence intervals.

Source (N)	Metric	W/L/S/U	Δ (pp)
Original (74)	Top-1	1/1/67/5	[−4.1, +5.4]
Original (74)	Top-5	0/0/64/10	[−9.5, +5.4]
Render (106)	Top-1	4/0/89/13	[−3.8, +12.3]
Render (106)	Top-5	1/1/82/22	[−10.4, +17.0]

Among feasible rules we maximize out-of-map rejection, then in-map acceptance, and break remaining ties in favor of fewer active gates and a lower threshold. We freeze these rules before scoring the 3,678 queries across nine scenes; the cohort has been used in earlier development and is not a new independent test set. Validation retention does not guarantee retention on these queries.

The raw match ratio divides RANSAC inliers by all saved raw matches; the pair ratio instead uses the deduplicated query-keypoint/landmark pairs supplied to PnP. A unique-keypoint ratio uses distinct inlier query keypoints over distinct matched query keypoints. For refined LoGS, final support is recomputed at the unchanged refined pose using positive depth and a 12-pixel reprojection bound. Initializer-score gates in that panel select already-refined poses and do not save refinement computation. SplatHLoc does not save a correspondence denominator, so we evaluate its observed final inlier count only.

On fixed initializer poses, recalibration selects a distinct count of 40. Relative to a raw inlier-count threshold of 40 on the same initializer, it removes 11 additional out-of-map returns and two incorrect in-map returns, without losing a correct pose. This geometric route is the QGVI-

Table S17. Calibrated RGB rejection controls. All entries pool the nine scenes (1,839 in-map and 1,839 out-of-map queries). FAR and FRR use the out-of-map and in-map denominators, respectively; pose success includes rejected in-map queries as failures. Precision is correct $1\,\mathrm{m}/2^\circ$ poses divided by all accepted queries, including out-of-map returns. $100U_1 = 100(S - W)/3678$. Best values are bold within each LoGS pose family, with ties included; boldface does not imply joint optimality across columns. The SplatHLoc row has its own native poses. † The added threshold is zero because native eligibility already requires four final inliers; thresholds zero and four are equivalent. All gates preserve retained pose coordinates.

Acceptance rule	Threshold	FAR↓ (%)	FRR↓ (%)	Pose success (%)↑ $1\,\mathrm{m}/2^\circ$	Pose success (%)↑ $5\,\mathrm{m}/5^\circ$	Precision↑ (%)	$100U_1$ ↑
Fixed LoGS initializer poses (no photometric refinement)							
Return every pose	—	100.000	**0.000**	**96.52**	**96.68**	48.26	-3.48
PnP success only	—	99.674	0.054	**96.52**	**96.68**	48.35	-3.29
Raw inlier count	40	1.305	1.033	**96.52**	**96.68**	96.26	46.38
Raw match inlier ratio	0.1	79.445	0.761	96.36	96.47	53.93	7.01
Pair inlier ratio	0.1	85.101	0.435	**96.52**	**96.68**	52.27	4.19
Unique-keypoint inlier ratio	0.2	24.035	0.979	96.41	96.57	78.35	34.88
Unique inlier count (refit)	40	**0.707**	1.142	**96.52**	**96.68**	**96.94**	**46.74**
QGVI-I	40	**0.707**	1.142	**96.52**	**96.68**	**96.94**	**46.74**
Native LoGS refined poses (up to 20 photometric steps)							
Initial raw inlier count	40	1.305	1.033	**96.52**	**96.68**	96.26	46.38
Initial raw match inlier ratio	0.1	79.445	0.761	96.36	96.47	53.93	7.01
Final pair inlier count	40	1.251	1.142	**96.52**	**96.68**	96.41	46.47
Final pair inlier ratio	0.1	81.185	**0.435**	**96.52**	**96.68**	53.40	6.14
QGVI-I + photo.	40	**0.544**	1.142	**96.52**	**96.68**	**97.10**	**46.82**
Final unique-keypoint ratio	0.2	22.023	0.979	96.41	96.57	79.65	35.89
Native SplatHLoc final poses							
Final inlier count	$0^\dagger$	99.347	0.272	61.17	65.47	30.73	-38.36

Table S18. Scene-level comparison of the calibrated distinct-keypoint gates. G denotes the geometric QGVI-I default; R denotes QGVI-I + photo. Both require at least 40 distinct inlier query keypoints at their returned pose and have identical pose-success and FRR entries in each scene. Precision and utility use the strict pose threshold. Bold highlights the better G/R value within each scene, including ties. Estimated DA3 scales for five scenes qualify the metre-based pose thresholds.

Scene	N_{in}	Pose success (%)↑ $1\,\mathrm{m}/2^\circ$	Pose success (%)↑ $5\,\mathrm{m}/5^\circ$	FRR (%)↓	FAR (%)↓ G	FAR (%)↓ R	Precision (%)↑ G	Precision (%)↑ R	$100U_1$ ↑ G	$100U_1$ ↑ R
Block Small	139	90.65	92.09	3.597	**0.000**	**0.000**	**94.03**	**94.03**	**42.45**	**42.45**
Block A	602	93.36	93.52	2.492	**0.332**	**0.332**	**95.42**	**95.42**	**44.44**	**44.44**
NYC	180	100.00	100.00	0.000	1.111	**0.556**	98.90	**99.45**	49.44	**49.72**
SF	174	100.00	100.00	0.000	4.023	**2.874**	96.13	**97.21**	47.99	**48.56**
City Sample	142	100.00	100.00	0.000	**1.408**	**1.408**	**98.61**	**98.61**	**49.30**	**49.30**
Colosseum	126	91.27	91.27	0.794	**0.000**	**0.000**	**92.00**	**92.00**	**41.67**	**41.67**
Elven Ruin	118	100.00	100.00	0.000	**0.000**	**0.000**	**100.00**	**100.00**	**50.00**	**50.00**
Park	228	100.00	100.00	0.000	**0.000**	**0.000**	**100.00**	**100.00**	**50.00**	**50.00**
Road	130	100.00	100.00	0.000	**0.000**	**0.000**	**100.00**	**100.00**	**50.00**	**50.00**

I default. The optional QGVI-I + photo. ablation applies the final distinct-keypoint gate to full LoGS and gives 10 false accepts instead of 13, at the same 21 in-map rejections and 1,775 strict successes. It uses native LoGS refinement, without a new pose optimizer. These comparisons use the exposed development cohort rather than an independent final test. For SplatHLoc, a threshold of eight would reject more negatives but lose one validation pose correct at the looser threshold; the no-loss criterion therefore selects its native acceptance point. This is a constraint on the chosen operating point, not evidence that SplatHLoc cannot reject.

Threshold-swept risk–coverage summaries qualify the fixed operating-point comparison. On the identical 3,671 PnP-eligible initializer poses, the grouped right-step area is 0.17638 for unique inlier count, 0.17492 for raw inlier count, and 0.16641 for the unique-keypoint ratio. The

unique count therefore has the best reported validation-constrained operating point, but not the lowest integrated risk. These areas cover the observed eligible range only (0–99.81% coverage), without breaking score ties; they are not compared across methods with different poses or eligibility.

Paired photometric diagnostic. We additionally compare the saved initializer with native LoGS refinement (up to 20 steps) for every RGB query. All 3,678 pose matrices change. Among the 1,839 in-map queries, strict success remains 1,775 and loose success remains 1,778, with zero individual gains or losses at either threshold. Continuous pose errors nevertheless change: median translation error increases from 0.01191 to 0.02757 m, and median rotation error from 0.02329° to 0.08531°. Translation error improves for 405 queries and worsens for 1,434; rotation error improves for 129 and worsens for 1,710 (a 10^{-9} tolerance in metres or degrees). These aggregate observations on the existing development cohort show no threshold-success gain from refinement and worse median errors, while QGVI-I + photo. has three fewer false accepts. QGVI-I retains the geometric initializer as its default; photometric refinement remains an optional ablation. These results do not establish a general benefit or failure of photometric refinement. Coordinate preservation applies to each acceptance gate, not to the photometric optimizer.

L. Fixed-Pose Acceptance Audit

Table S19 isolates rejection from pose estimation in QGVI-I, the geometric default. Every retained camera matrix is unchanged, and every initializer pose correct at either threshold remains accepted on this cohort. The precision gain therefore comes from removing incorrect outputs, including out-of-map returns, rather than improving pose coordinates.

M. Historical Text Retrieval and Verification Controls

QGVI-T here denotes historical fusion.

On 176 queries, verification yields 148–155 Top-5 successes with captions and 134–144 with CLIP, versus 142–175 for captions alone (Table S20); overlapping bounds preclude a verifier-effect claim. Verification gives 45/180 empty returns. Counting is untested: 30/34 descriptions lack target counts (Table S21).

Acknowledgments

ChatGPT (OpenAI) assisted with manuscript drafting and revision, code development, and experimental workflow support. The authors are responsible for the methods, results, and conclusions.

Table S19. **Selective acceptance at fixed poses.** The two arms use the same LoGS geometric estimates as QGVI-I, without photometric refinement, and are evaluated on 3,678 development queries; the ungated arm includes seven retrieval fallbacks. Pose success uses all 1,839 in-map queries; precision uses all accepted outputs, including distractors. Pooled percentages; the gate changes acceptance only. Bold marks better rates and ties.

Metric	Ungated	Gated
Accepted outputs (n)	3,678	1,831
False accepts / out-of-map	1,839/1,839	13/1,839
False accept rate ↓	100.000	**0.707**
False rejects / in-map	0/1,839	21/1,839
False reject rate ↓	**0.000**	1.142
Pose success, 1 m/2° ↑	**96.52**	**96.52**
Pose success, 5 m/5° ↑	**96.68**	**96.68**
Precision, 1 m/2° ↑	48.26	**96.94**
Precision, 5 m/5° ↑	48.34	**97.11**

Table S20. **Text component ablations.** (a) Pooled relevance-only Top-5 bounds (%) for 176 DeepSeek-authored/judged development queries. (b) Equal-scene joint Top-1/Top-5 success for 180 queries: blind Qwen3-VL-8B-Instruct relevance and DeepSeek validity reviews; source support reviewed separately. Bold marks maximal endpoints (a) or success (b); overlapping bounds preclude ranking.

(a) Retrieval and verification

Retrieval	Visual verifier	n	Relevant Top-5 success (%) ↑
CLIP	No	176	68.2–97.2
CLIP	Yes	176	76.1–81.8
Captions (QGVI-T)	No	176	80.7–**99.4**
Captions	Yes	176	**84.1**–88.1

(b) Joint relevance and validity on the 180-query cohort

Method	Top-1 (%) ↑	Top-5 (%) ↑
QGVI-T	58.3	**78.3**
QGVI-T + verification	**66.1**	72.8
CityLoC, without refinement	25.0	30.7
CityLoC, with refinement	26.1	34.6

Table S21. **Scene and query-type breakdown.** Relevance-only Top-5 bounds (%) for 176 exposed DeepSeek-judged queries (n: unique IDs). Author-assigned types lack independent source grounding; answerability/validity are unverified. Bold marks each row's maximal endpoints, including ties; overlapping bounds preclude ranking.

Scene / query type	n	CLIP + verifier Top-5 (%) ↑	QGVI-T + verification Top-5 (%) ↑
Block Small	20	**85.0**	**85.0**
Block A	20	80.0–**85.0**	**85.0**
NYC	20	**90.0**	**90.0**
SF	20	80.0–95.0	**85.0–100.0**
Citysample	20	65.0–70.0	**80.0**
Colosseum	16	75.0–87.5	**87.5–100.0**
Elvenruin	20	60.0–75.0	**75.0–85.0**
Park	20	90.0	**95.0**
Road	20	60.0	**75.0**
Object	61	88.5	**91.8–95.1**
Attribute	35	80.0–85.7	**91.4**
Spatial relation	35	71.4	**85.7**
Counting	34	64.7–**88.2**	**73.5–88.2**
Visible text	11	**45.5**	**45.5**
All queries	176	76.1–81.8	**84.1–88.1**